

SITE SAFETY PLUS

Construction site health and safety

C: General safety



GE700C
OFFICIAL CITB
PUBLICATION
Version 10.1

Published by CITB, Sand Martin House, Bittern Way, Peterborough, PE2 8TY

© **Construction Industry Training Board 1982**

First published in 1982 as GE700 (A4 ring binder x2)

Republished in 2013 as GE700C (A4 book)

Revised 2014, 2015

Reprinted Jan 2016

Revised 2017

Reprinted Dec 2017

Revised Dec 2018, Dec 2019, Dec 2020, 2022

Revised 2023, 2024

ISBN 978-1-85751-583-1

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Supporting INFORMATION

GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



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Overview

In this chapter we provide practical guidance on factors to be considered when preparing to set up a construction site. Details on the other sections of GE700 that will support this work, including information for security, logistics, safety, signs, access and egress, and the environment are also referenced.

Generally, much of construction work is transient and therefore the hazards on site can change from day to day. Many accidents occur on site as a result of poor design and planning during the pre-construction phase.

1.1 Introduction

This chapter aims to give employers and managers general practical guidance on site set-up, organisation and layout, how a site should be run and its impact on site workers, people living nearby and visitors. The chapter also introduces some of the topics found in other sections of GE700. Its content should be read alongside, not instead of, those sections.



The construction phase plan must record health and safety arrangements for the construction phase, site rules and measures to control particular risks (such as work near high-voltage cables, risk of drowning and working below ground, for example, working in excavations).



When planning site set-up and logistics, use site drawings to outline (or use scale cut-outs to lay out) site accommodation, plant and vehicle turning circles, material storage, buildings and temporary works (such as scaffold) and excavation working space. Also, when planning the safe management of deliveries to and from the site, consider the proximity of the site to nearby housing, schools, hospitals or shopping centres. Remember to take into account the finished building or structure when decommissioning the site.



It may be helpful to use a sheet of clear Perspex over a site plan that outlines items such as site accommodation, fencing and temporary roads. Work (such as excavations, crane positions and material storage) can then be colour coded and plotted daily, making it easier to identify and plan for potential logistical and safety issues.

The **Construction (Design and Management) Regulations 2015** (CDM) place specific duties on commercial clients, the contractor or the principal contractor to ensure that authorised personnel have access to appropriate information to keep them safe on site. In addition, these duty holders have a duty to ensure that unauthorised persons do not gain access to site. Unauthorised persons will not be familiar with site rules and site hazards, so access to site could expose them to hazards that they may not be aware of.



For further information on what to do when setting up and managing a site refer to the CDM industry guidance documents for contractors and principal contractors.

In accordance with the **Occupiers' Liability Act 1957**, trespassers and non-employees have a right to be protected from risk if entering a site. A higher standard of care must be given by site management where children are concerned, for both authorised and unauthorised access.

1.2 Important points

The term *site organisation* generally includes the following areas.

- Management of the site.
- Preparation and communication of information.
- Pre-job briefing.
- Competence.
- Site discipline.
- Welfare requirements.
- Site security.
- Safe storage of materials.
- Safety signage.
- Utilities.
- Communication with neighbours.

1.3 Considerations when setting up a construction site

1.3.1 Existing site features

- Consider previous uses of the site, including contaminated ground, underground tanks or unexploded ordnance that may need to be cleared. Also consider any previous storage of dangerous goods or hazardous substances in buildings, cellars or tanks, which might require substantial decontamination, resulting in controlled site transport movements and additional time and cost.
- Check if any existing health and safety files or historic records are available detailing any hazardous previous use of existing buildings or structures on site.
- Check that an asbestos survey has been carried out to identify areas that need protection or isolation.
- Consider whether demolition is required as part of the project.
- Confirm if any underground services need to be isolated or if barriers and goalposts are required to avoid contact with live overhead power lines. Consider the impact on existing, or planned, access routes on site.
- Are there any protected wildlife or plants?
- Check if any invasive species (such as Japanese knotweed) are present.
- Are there watercourses (underground or surface) that need to be protected from contamination?
- Are there archaeological features that need protection?
- Check neighbouring activities to identify sensitive receptors and plan to avoid nuisance.



For further information refer to Chapter D07 Underground and overhead services and Chapter B09 Asbestos.

1.3.2 Site security

A principal contractor – or, where there is only one contractor, the contractor – must take the necessary steps to prevent access to the site by unauthorised persons or trespassers.

In accordance with the Occupiers' Liability Act, trespassers and non-employees have a right not to be put at risk if they enter a construction site.

The contractor must not start work until the following steps have been taken, either by themselves (on a one-contractor project) or by the principal contractor (on a project involving more than one contractor).

- Secure the boundary immediately on possession, preferably with lockable gates and suitable hoarding to prevent unauthorised access or fly-tipping.
 - The perimeter of a construction site must be identified by suitable signs, so it can be recognised, and it should be fenced off or secured where practicable.
 - Consider installing closed-circuit television (CCTV) cameras. Even on small sites this can prevent or deter unauthorised entry and can support other HSE monitoring.
 - Any hoarding or barriers provided to maintain site security and control unrestricted access must be checked regularly to ensure that they remain safe throughout the duration of the construction project.
 - If the whole site cannot be secured then ensure areas with potentially hazardous operations are appropriately secured.
 - Make specific security arrangements for particular areas (such as hazardous substances or plant and equipment storage).
 - On larger sites, access for site personnel may be controlled by a biometric system, which can provide a solution for ensuring site security. Biometric verification is any means by which a person can be uniquely identified, evaluating one or more distinguishing biological traits, such as fingerprint, retinal scan, or facial recognition.
 - Make provision for card reading facilities to identify individuals and confirm competency (often the CSCS card reader).
- In planning for site security, remember that children often think that construction sites are exciting places to play. It is therefore important that access to the site is controlled during and after normal working hours.
- Any evidence of trespass must be reported to site management immediately and appropriate action should be taken.



A facial recognition system for controlling access to a site



Seven year old boy trapped and suffocated on construction site

A civil engineering firm was fined £600,000 for safety breaches after a seven-year-old child became trapped and suffocated on a construction site.

The boy went missing from home and was found the next morning by workers at the construction site in South Yorkshire.

An investigation by the Health and Safety Executive (HSE) found that he had become trapped in a drainage pipe, which had been fixed into the ground in preparation for the installation of fencing posts. Tragically, he had suffocated before being found the next morning when work restarted on site.

Howard Civil Engineering Ltd pleaded guilty to breaching regulation 13(4)(b) of the Construction (Design and Management) Regulations 2015 and to breaching Section 3 (1) of the Health and Safety at Work etc. Act 1974. The company was fined £600,000 and ordered to pay £42,952.88 in costs at Sheffield Crown Court.

The construction site was a new-build housing development next to an existing housing estate and adjacent to busy pedestrian footpaths and roads. The HSE found that there was insufficient fencing in place to prevent unauthorised persons from accessing the construction site due to a combination of poor planning, management and monitoring of the site and its perimeter.

Speaking after the hearing, the HSE inspector said: "The child should never have been able to be on that site. He should have been kept out. The construction industry should be aware of the dangers of construction sites to members of the public and any other unauthorised persons.

"The dangers to children gaining access to construction sites and treating them like a playground is an ongoing problem which must be addressed at all types of sites no matter what their complexity or size.

"The industry must do all it can to ensure children can't access construction sites and be exposed to the inherent risks they present to prevent further tragedies like this from occurring."

(Source: HSE.)

1.3.3 Temporary site structures

A competent person must ensure the following.

- They must carry out a foundation assessment and check whether the area for site cabins or other temporary structures is suitable.
- Assess temporary site roads for their capability to withstand loads during construction works, especially if they cross existing services.
- Undertake checks of the ground-bearing capacity of any hard standing for the positioning of lifting equipment.
- Ensure that material storage areas are planned carefully, with an awareness of shallow services that could be damaged.
- Design the site hoardings and gates to withstand local wind loadings, particularly to ensure swing gates are managed during high wind conditions. The Temporary Works forum can provide additional information about other conditions for site hoardings.
- If a 110-volt distribution system is required, check if a live 230/400 volt system is available to power it. Check who will design and install it, and who will test it and ensure that it meets ECA/NICEIC standards.



For further information refer to Chapter C04 Temporary works.

1.3.4 Pedestrian and traffic routes

- Ensure safe vehicle operations and safe management of deliveries are considered at the design stage of a project, by specifying suitable profiles and removing hazardous gradients and embankments.
- Traffic and pedestrian management should be planned and then covered during the site induction. A plan of the construction site and traffic management plan could also be provided at the entrance for visitors and delivery drivers.
- Minimise risk to pedestrians by providing separate routes and entrances for vehicles and pedestrians. Use physical segregation where possible.
- Ensure the safety of the public on a partially-completed housing development site, by having phased occupation of dwellings: so that site traffic can be excluded from occupied areas, and public vehicle and pedestrian routes can be segregated from site vehicle and site worker routes.
- Ensure site entrances and exits minimise traffic hazards on public roads.
- Any traffic route must not be so uneven, potholed, sloped or slippery that any person using it is exposed to a risk to their safety.
- Provide a car park for privately owned vehicles, preferably accessed directly from a public road, with vehicle entry to the construction area prevented, if practicable.

- Parking areas should be clearly signposted, firm, level or with an acceptable gradient, well lit and not slippery.
- Ensure that site offices and other facilities are connected to the parking areas or pedestrian entrance, avoiding the need for visitors to cross site to gain access.
- Ensure safe, designated pedestrian routes, separate from vehicle routes where possible, are available to all work locations and are maintained as work progresses.
- Ensure pedestrian routes are wide enough to accommodate the number of people likely to use them at peak times.
- Provide pedestrians with a clear view of traffic movements at crossing points.
- Make use of a turning area if space is tight and try to avoid the need for reversing on site – one way systems are ideal.
- Set standards for driver competence and vehicle safety.
- Ensure availability of competent vehicle marshallers if reversing is required, entry onto a site is awkward, or pedestrians cannot be completely segregated.
- Roads should be wide enough for the vehicles using them.
- Sharp and blind bends should be avoided. Additional control measures (such as mirrors, stop signs or signals) could be used to help drivers and pedestrians see what is around the corner.
- Consider the possible requirement for a wheel wash and and/or a road sweeper to keep site and public roads clear of mud and other debris, avoiding the creation of hazardous driving conditions or nuisance.
- Reduce the likelihood of pedestrian injury in areas of ongoing work with suitable temporary control measures.



The safe separation of construction plant and pedestrians avoids accidents



Access from car park directly to site offices



Pedestrians have a clear view of traffic movements at crossing points



WISA block impalement protection in use on steel reinforcing bars

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1.3.5 External and internal site access routes and housekeeping

- Access routes should be on an even surface, signposted and have adequate lighting.
- If a route becomes muddy or slippery it should be treated with hardcore or a similar material.
- If internal access routes are icy, they should be treated with grit or given a temporary covering to make them safe to walk on.
- Careful consideration should be given to location and routing of temporary electrical supplies to minimise trip hazards and protect electrical supply cables from damage.
- Maintain high standards of housekeeping to keep all access routes clear, reduce tripping and fire hazards and project a positive company image.
- All obstructions should be kept out of access routes. This can be achieved by providing designated areas for deliveries and waste.



Good housekeeping – cables routed along ceiling and over doors to minimise slips and trips

1.3.6 Site lighting

The Provision and Use of Work Equipment Regulations 1998 require that:

“ Every employer shall ensure that suitable and sufficient lighting is provided at any workplace.

CDM require that adequate lighting is provided at every place of (construction) work and approach to the workplace, and that secondary (back-up) lighting is provided where failure of the primary lighting would result in risks to health or safety.

These two legislative provisions place a duty on employers to ensure that well-planned and adequate lighting be installed on site where necessary for safe working and access.

To obtain such lighting conditions, you should consider the following.

- Installing suitable lighting to provide the required level of illumination for the nature of the work being carried out.
- Mounting the lights at a suitable height above the work level to give the required spread of light.
- Positioning lights to avoid glare, dazzle and reflection.
- Changing the position of lights as work proceeds.
- Screening or shielding lights from reflective surfaces, on traffic routes, neighbouring properties and so on.
- Routing of lighting cables to avoid trip hazards or damage to the cable.
- Installing back-up lighting if failure of the primary sources would result in risks to health and safety.

“ Certain forms of lighting can constitute a heat source and, therefore, a fire hazard. They also retain heat for a period after being switched off. Avoid locating lights in close proximity to combustible materials very employer shall ensure that suitable and sufficient lighting is provided at any workplace. Halogen floodlights get particularly hot, and have been the cause of several fires.

1.3.7 Material unloading and loading

The loading and unloading of vehicles has been the cause of a number of fatalities. Drivers and operatives unloading cargoes have been hit and crushed by objects falling from the vehicles because of the following circumstances.

- Cargo moving during transit and not being securely held.
- Failure of wagon straps, resulting in the release of loads suddenly and without warning.
- Changes in load distributions leading to uncontrolled collapse of the loads.

In addition, access and falls from height from the flat bed of a wagon need to be controlled.

The following measures should be considered.

- Loading stations with fixed platforms.
- Vehicle based systems (such as handrails).
- Soft-landing systems placed around the wagon.

- Inertia harnesses.
- No-slip surfaces.
- Keeping the area clear of debris and rubbish.
- Unloading checks to confirm the integrity of the retaining mechanisms.
- Checks to confirm the locations of lifting points.
- Checks to confirm the load's centre of gravity.

Reduce the risk of accidents involving mobile plant and vehicles by designing groundworks and landscaping that reduce the need to import fill or take spoil off site.

Materials that are unloaded or tipped from plant should be deposited in designated locations where they will not cause an obstruction, environmental issues or a hazard.

A vehicle marshaller should assist the driver during any manoeuvring or tipping operation to ensure that it is safe to tip and that there are no other hazards to be encountered during the operation.

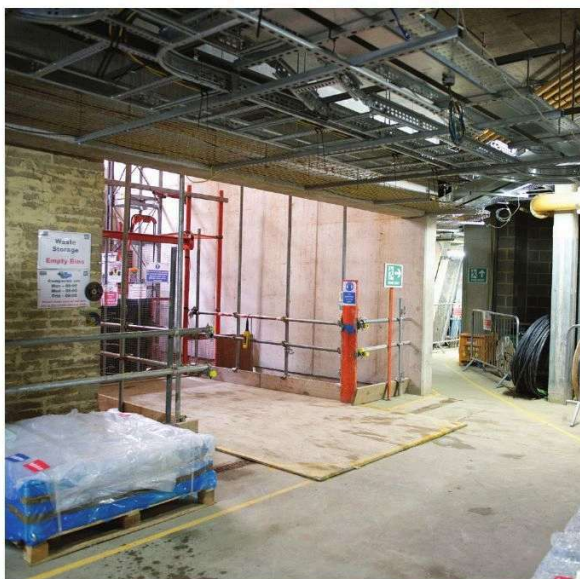
Consideration should also be given to additional precautions that may be required (for example, increased signage, entry and exit signs into and out of delivery areas, restricted pedestrian movement, proximity sensors and/or convex mirrors).

Ensure that any plant being used is in good condition, is appropriate for the task and is operated by competent, certified employees.

Operatives working with plant or vehicles, or on sites where there are regular vehicle movements, should be provided with the appropriate personal protective equipment (PPE), which should include high-visibility clothing.

1.3.8 Materials storage and lay-down areas

- Develop a materials delivery schedule and ensure that it is communicated to all relevant personnel.
- Ensure all contractors and suppliers are aware of any delivery issues or restrictions specific to the site or surrounding area (such as schools, low bridges or one-way systems).
- Make provision for designated and safe storage areas to protect materials and ensure site and access roads are kept clear of obstructions.
- Make provision for safe and secure storage of any hazardous, flammable and polluting substances (such as bottled gases, oils and fuels).
- Make sure pallets of materials are stored in stable stacks with a safe limit designated for each different material.
- Consider what access is needed to reach these materials and what lifting equipment may be required (such as telehandlers, forklift trucks, mobile or tower cranes).
- Consider ways of moving materials on site without using vehicles.
- Try to place heavy items that need to be moved by hand near to where they are required, to minimise manual handling and avoid double handling.



Marking access routes and providing designated storage areas helps keep walkways tidy and free from obstruction



Designated plant accessory storage area



Supervisor convicted of manslaughter after a member of the public is crushed and killed

A 43-year-old woman was walking along a public pavement past a central London construction site when three large, unglazed window frames, weighing 655 kgs, fell and killed her. The window frames had been delivered the previous day and had been left on the public pavement overnight, leaning against the building. No efforts had been made to secure them to prevent them from falling and nor had any barrier been placed around them to protect members of the public using this busy thoroughfare.

The construction site was under the control of Westgreen Construction Limited (Westgreen), which was the principal contractor on a project to refurbish the ground floor and basement of the building to create a gallery space on behalf of the client. This included installing new ground floor windows and doors.

By the time of the accident, the construction phase of the project was well under way but it was running behind schedule. As the principal contractor, Westgreen was responsible for planning, managing and monitoring the construction phase of the project to ensure, so far as was reasonably practicable, that it was carried out without endangering those working on the project or members of the public. On at least one occasion in the days before the accident, a plywood hoarding had fallen from one of the apertures on the building, almost hitting a member of the public.

After identifying a catalogue of failures which led to the accident, the following convictions were secured.

- A construction supervisor was convicted of manslaughter (12 months' imprisonment) and breaching Section 7 of the Health and Safety at Work etc. Act 1974 (nine months' imprisonment). The supervisor was employed by IS Europe Limited, which was contracted by Drawn Metal Limited, to install the window frames.
- The site manager, working for the principal contractor Westgreen, was sentenced to six months' imprisonment, suspended for two years. Westgreen contracted Drawn Metal Limited to manufacture, deliver and install the window frames.

Failures

- Poor assessment of contractor competence.
- Non-compliance with an established construction phase plan and safe systems of work.
- No lifting plans in place for the operation.
- No procedures for checking risk assessments or method statements.
- Failure in the identification of hazards and mitigation of risk.
- Unsuitable, insufficient and out-of-date risk assessments and method statements.
- Contravention of existing risk assessments and method statements.
- Inadequate levels of communication and co-operation.
- Failure to postpone or refuse the delivery of the frames.
- Inappropriate storage of materials.
- No segregation between members of the public and construction activities.
- No segregation between members of the public and stored construction materials.
- Unsuitable storage and security of materials on site.
- Failure in monitoring the activities of sub-contractors working on site.

Following the verdict, the deceased woman's parents said: '[Our daughter] was a bright, lovely, professional woman living her life to the full and making plans for the future. Her future was taken from her when she was crushed to death by half-tonne window frames which took two seconds to fall on her. The frames had been left standing, almost vertically, at the side of a public pavement, unsecured to anything, unattended and with no safety barriers around them. If construction companies and the people who work for them are not held to account for such high levels of negligence and incompetence, then none of us is safe walking the streets next to construction sites.'

(Source: Homicide and Serious Crime Command, Metropolitan Police)



Materials must be stacked and secured appropriately to avoid them falling and injuring someone.



For further information visit the HSE website for *A guide to workplace transport safety* (HSG136).

For further information refer to:



- Chapter C06 Mobile work equipment
- Chapter C07 Lifting operations
- Chapter C08 Lifting equipment.

1.3.9 Utility services

Give early consideration to the following points.

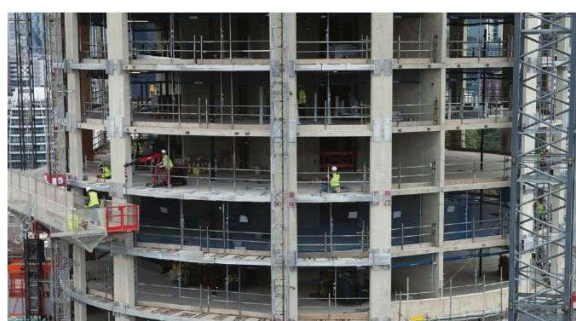
- The provision of a 230 or 400-volt three-phase (mains) supply when there may be heavy electrical loads (for example, a tower crane).
- The supply of clean water, which must be suitable for drinking, and may require installation and commissioning by the local water company.
- The drainage of surface and foul water. The Local Authority needs to be aware of any new connections and discharge of waste effluents into foul sewers. Discharge consents are required if water pumped from excavations or site run-off are discharged into rivers or other watercourses.
- The provision of telephone and/or broadband connections.
- The provision of gas or oil as required for heating.
- The size of the site and emergency requirements to determine what communication links are needed.
- The provision of an appropriately sized generator (if required) and supply of fuel.
- The incorporation of renewable energy and rainwater harvesting systems.



For further information refer to Section E: Environment.

1.3.10 Work at height

- A comprehensive risk assessment will determine the most appropriate type of equipment required.
- Consider methods of fall prevention on or around vehicles for the unloading of site resources (such as accommodation units and site fencing).
- Temporary site accommodation, welfare units and containers can be pre-slung or slung using a method that attaches to the bottom of the unit's legs rather than having to access the roof to attach slings to the top of it.
- If scaffolding is being used, it must be erected, altered or dismantled by, or under the direct supervision of, a competent person and be inspected periodically by a competent person.
- If mobile elevating work platforms are used, they must be suitable and operated by competent persons. Ground conditions need to be checked and proximity hazards (such as overhead cables, rivers and basements) must also be identified and controlled.



Various working at height activities



For more information refer to Chapters C07 Lifting operations and D01 Work at Height Regulations.

1.3.11 Excavations

- Sufficient working space for spoil heaps, earthwork support, laydown areas, pipes, and plant and machinery must be considered when planning the work.
- Suitable equipment (such as guardrails) will be required to form effective, physical barriers around excavations to prevent falls of persons, machinery or vehicles.
- Provide suitably anchored stop blocks or similar devices to prevent vehicles, especially tipping dumpers, from falling into excavations.
- If sides of an excavation cannot be battered, then they must be supported to prevent collapse of soil, rock or materials. A system of support must be designed by a competent person.
- Ensure safe means of access and egress for persons who need to enter the excavation; this should be by a secured ladder, or using adjustable lightweight staircase or towers.
- Never use underground service lines as a stepping point to get into and out of an excavation.
- Arrangements must be in place for statutory inspections of all excavations.
- Provide equipment to facilitate the evacuation of injured personnel.
- Give consideration to groundwater removal from excavations.



Physical barrier around an excavation



For further information refer to Chapter D06 Excavations.

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1.3.12 Site office

- If the site office has more than one level, particularly where non-construction staff are employed, then temporary planning permission may be required. Also check for any planning restrictions in the contract documents at this stage.
- When temporary accommodation is required, consider its environmental performance and select on the basis of high insulation, double glazing, door closers and PIR (passive infrared sensor) lighting.
- Administrative staff and others, who may not have PPE, should be provided with safe access to the site office without having to pass through operational areas. If this cannot be achieved at all times then they must be issued with appropriate PPE.
- Site offices should offer comfortable accommodation for all types of weather and ranges of temperature.
- Provide a suitable and safe form of heating.
- Provide a suitable and safe electrical supply and telecommunications.
- Compile registers for portable appliance testing, fire alarms and other emergency planning.
- Ensure the site visitors' log is available at the entrance to the site, is completed by all visitors on arrival and departure and is monitored.

1.3.13 Welfare facilities

- It is a legal requirement to provide clean and hygienic areas, where workers can securely store personal items and dry clothing, get changed, make a hot drink, heat and eat meals and take shelter in bad weather.
- Suitable provision must be made for toilets, washing (if necessary, shower) facilities, changing rooms and rest facilities. Where there are both women and men on site, separate conveniences must be provided, with doors capable of being secured from the inside.
- It may be necessary to provide alternative temporary arrangements until the welfare facilities, power or water connections are made. A variety of standalone, self-contained units are available to suit site-specific needs.
- Additional welfare facilities may be needed for peak workforce levels and provision should be made for their delivery and installation.



Facility for heating food and with access to running hot water



Separate male and female site toilets



For further information refer to Chapter B02 Welfare facilities.

1.3.14 First aid

- The appropriate number of appointed persons, emergency first aiders or qualified first aiders will be needed, depending on the number of people on site and the findings of the first-aid needs assessment.

As a guide:

- **fewer than five** – at least one appointed person (person responsible for the first-aid kit and calling the emergency services)
 - **five to 50** – at least one person trained in emergency first aid at work (EFAW) or first aid at work (FAW), depending on the type of injuries that may occur
 - **more than 50** – at least one person trained in FAW for every 50 people employed (or part of 50. For example, if there are 51 people on site, two persons trained in FAW will be required).
- The following information must be displayed and communicated, and should be covered as a topic during site induction.
 - The details of people who are first aiders, and the means of identifying them.
 - The location of first-aid kits, and the means of summoning first aid and medical assistance.

- Make sure the first-aid equipment has been ordered.
- Make sure appropriate signs are in place to identify first-aid locations and to direct personnel to them.
- Make sure you have the contact numbers and directions for emergency services, local hospitals and doctors.



For further information refer to Chapter B05 First aid.

1.3.15 Fire safety

- Gather all relevant pre-construction information, including any existing general fire precautions, building layout and the presence of flammable or combustible materials
 - Contact the emergency services to inform them of your location, the means of access and any planned specialist activities.
 - Work with the emergency services to agree site layout and identify areas and routes to be kept clear for emergency vehicle access.
 - Complete a fire risk assessment and develop this into a fire safety plan for the site.
 - Think about the level of risk from work activities and flammable substances that may be stored and used during the project.
 - Provide a sufficient number of fire extinguishers, suitable for the types of fire that might occur, and place them in highly visible fire points in appropriate places, including offices and welfare accommodation.
 - Train a suitable number of staff in the use of fire extinguishers.
 - Keep fire exits and emergency escape routes clear of obstructions. Make sure they are easily visible and have adequate, illuminated signage.
 - Ensure routes to the emergency assembly point are clearly indicated. The assembly point will be in a safe area, ideally outside of the confines of the construction site.
 - Maintain any third-party fire arrangements (for example, working in shared or occupied premises). Close liaison is vital.
- Be aware that the points above need to be reviewed regularly, given the changing nature of construction sites.



Emergency routes should be clearly marked and indicate the direction to the nearest place of safety



For further information refer to Chapter C02 Fire prevention and control.

1.3.16 Emergency plans

- From the start of any project, plans must be put in place for the safe evacuation of the site.
- On complex or refurbishment projects, it is good practice to liaise with local fire authorities to identify fire-fighting strategies for the plan, together with the locations of fire-fighting cores, dry risers and lifts or hoists to be used or not used.
- Emergency escape routes must be identified and labelled with appropriate signs and communicated to everyone on site.
- A suitable assembly point should be identified. This should accommodate the maximum number of people on site at any time, be a safe distance from potential hazards and not be an obstruction or hazard to the emergency services.
- Display information about actions to be taken in an emergency, including what the alarm sounds like, days and times when it will be tested and where the assembly point(s) are.
- For larger sites, the emergency plan may be part of the construction phase plan. Fire wardens may need to be appointed for specific areas of the site. Emergency plans should be covered as part of any site induction programme.
- Identify the responsible person, who will carry out a head count of people, to account for the people registered as being on site (this may be more than one person on large sites).
- Ensure that Environment Agency contact details are available and that suitable equipment is provided for use in the event of an environmental emergency (such as spill kits for use with a fuel spillage).
- It is important that workers are trained in the use of spill kits, rescue lines or other emergency equipment provided.

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1.3.17 Contact with emergency services

- Ensure the local emergency services are informed about the site's location, particularly if it is difficult to find.
- Provide details on the means of access and any specialist activities being undertaken.
- If you are running a large site, you should nominate an emergency controller to liaise with the emergency services and oversee site emergency actions and any communications with the press or neighbours.



Display a road map in the site office, showing the quickest route to the nearest hospital and police station. This information should also be included in the site induction.



Valuable time can be wasted if the emergency services (police, fire or ambulance) are not aware of your location

1.3.18 Personal protective equipment

- Employees (including agency workers if they are legally regarded as employees) must be supplied with appropriate PPE free of charge. The site management team should ensure that workers understand how, where and when to use it, what to do if it is lost or damaged and how to care for and maintain it.
- A stock of relevant PPE will need to be available on site, to cover specific tasks undertaken by employees, to replace damaged PPE and to issue to site visitors.
- For specialist work you may need to seek advice when selecting respiratory protective equipment (RPE), eye protection, hearing protection, gloves and some other forms of PPE.
- There should always be adequate means to clean, maintain and store PPE.



For further information refer to Chapter B06 Personal protective equipment.

1.4 Safety signs

All safety signs are colour coded, as required by the Health and Safety (Signs and Signals) Regulations 1996.

There are five types of sign, shown below.

- Prohibition.
- Warning.
- Mandatory.
- Emergency escape and first aid.
- Fire-fighting.

1.4.1 Prohibition signs

Prohibition signs are round, with a black pictogram on a white background, with red edging and a diagonal red line. They prohibit behaviour likely to increase or cause danger by informing people of things they **must not do**.



No smoking



No naked flames



No access for pedestrians



Do not extinguish with water



Not drinkable



No mobile phones



No access for industrial vehicles



Do not touch

1.4.2 Warning signs

Warning signs are triangular, with a black pictogram on a yellow background, and black edging. They warn people of a hazard or danger.



Combustible or flammable material



Explosive material



Toxic material



Corrosive material



Radioactive material



Overhead load



Industrial vehicles operating



High voltage



General sign (where necessary, accompanied by another sign)



1.4.3 Mandatory signs

These are round, with a white pictogram on a blue background. They prescribe specific behaviours by informing people what they **must do**.



Protective eyewear must be worn



Safety helmet must be worn



Hearing protection must be worn



Respiratory equipment must be worn



Safety boots/shoes must be worn



Safety gloves must be worn



Safety overalls must be worn



Protective face shield must be worn



Safety harness must be worn



Pedestrians must use this route



General sign (where necessary, accompanied by another sign)

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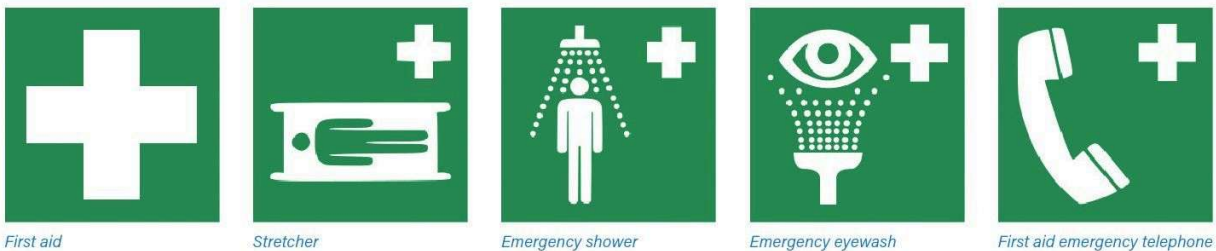
1.4.4 Emergency escape and first-aid signs

These signs are rectangular or square, with a white pictogram on a green background. They give information on emergency exits, assembly points and first-aid facilities.

1.4.4.1 Emergency escape route and assembly point signs



1.4.4.2 First-aid signs



First aid

Stretcher

Emergency shower

Emergency eyewash

First aid emergency telephone

1.4.5 Firefighting signs

These signs are rectangular or square, with a white pictogram on a red background. They indicate the location of firefighting and safety equipment.



Fire hose reel

Fire ladder

Fire-fighting equipment

Fire emergency telephone

Fire extinguisher

Fire alarm call point

1.4.6 Signs and notices

The Health and Safety (Safety Signs and Signals) Regulations 1996 require employers to provide safety signs where other methods, properly considered, cannot deal satisfactorily with certain risks and the use of a sign will further reduce that risk. Employers should ensure that their employees are aware of, and understand the meaning of, safety signs and signals, providing training where necessary.

- Signs should be displayed at the site entrance(s) to inform all persons entering the site about the health and safety requirements for the site. These should include, for example, minimum standards of PPE and state that all visitors must report to the site office.
- Other signs to be displayed include a selection of mandatory, warning, prohibition, emergency escape and first-aid signs.
- Ensure there is adequate fire signage (such as signs showing escape routes, fire assembly points and where the fire alarm call points (alarms and extinguishers) are located).
- Signs will also be required to identify the presence of liquefied petroleum gas (LPG), fuels and other flammable substances.

- Some signs may be used together with other types of communication, for example:
 - illuminated signs and acoustic signals
 - illuminated signs and verbal communication
 - hand signals and verbal communication.
- Traffic management signs will also be required and should meet the requirements of the latest (road) Traffic Signs Regulations and General Directions 2016, as appropriate to the specific site. These may also include directional signage to site and instructions for deliveries.



Signs for workers to use, to highlight hazards as working environments change



Signs should not be used as the only means to convey important information. They should be used to reinforce the information you have already covered in a site induction.

1.5 Environmental considerations

Construction work can have a severe impact on the environment. Therefore, measures must be taken to prevent or reduce the potential for harm or nuisance.

The risks should be recorded in the environmental impact assessment (EIA) and the construction environmental management plan (CEMP). Review the CEMP, as follows.

- Identify and protect watercourses and drainage systems.
- Ensure you have licences for discharges of water or effluents and arrange for settlement tanks, if appropriate.
- Identify, enhance and protect habitats and areas occupied by protected species of animals, insects and plants.
- Identify and protect areas of archaeological and historic importance.
- Create an action plan detailing measures to be taken in the event of a spillage to prevent ground or watercourse contamination. Include emergency contact details for the relevant Environment Agency.
- Identify how you propose to reduce noise and dust levels, as far as reasonably practicable. You may need to consider restricting some construction processes to certain times of the day.
- Ensure correct handling, storage and disposal of hazardous and non-hazardous waste.
- Avoid nuisance to neighbouring properties by locating noisy activities away from boundaries. Ban the use of radios, do not leave engines running when not in use and avoid inconsiderate parking on public highways.
- Clearly identify and explain environmental issues in inductions and briefings.
- Consider methods of communicating with the local community to keep them informed about site activities that may have an impact on them.
- Demonstrate a caring attitude towards local residents with regard to site lighting and CCTV use during the hours of darkness.
- Consider electing an environmental champion for the site.



For further information refer to Section E: Environment.

1.6 Waste storage and disposal

If you are involved in managing waste, you have a legal duty of care.

The duty of care applies to everyone involved in handling the waste, from the person who produces it to the person who finally recovers or disposes of it.

Waste that is created on a site must be disposed of appropriately by licensed contractors. Hazardous waste will need to be stored in appropriate containers and removed by contractors with the appropriate hazardous waste licence (special waste in Scotland), with the transfer recorded using appropriate waste consignment notes.

A plan to reduce and manage waste should be developed before the project commences. Relevant content should also be communicated to suppliers, contractors and those carrying out the work.

The following will need to be considered.

- Will the project be carrying out any waste treatment activities requiring permits or exemptions?
- Can any treated waste be reused on site or transported for use on a registered cluster site (a site that utilises surplus materials)?
- Is the waste suitable for transporting (hazardous or non-hazardous) and does it require special handling by competent persons?

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- All waste must be transported by a registered waste carrier and be supported by a waste transfer note or, in the case of hazardous waste, a consignment note (special waste in Scotland) to an authorised facility.
- The provision of an adequate number of suitable, clearly marked skips or secure space to allow waste (for example, plasterboard, metal, timber and electrical waste) to be segregated prior to removal. This can help improve recycling and reuse, and reduce the amount of waste going to landfill.



For further information refer to Chapter E10 Waste and material management.



A waste chute and tipping skip provide a safe and easy method of collecting and removing waste from under a scaffold (ideally, the chute and skip would be shrouded to reduce dust generation and escape).

1.7 Human factors

1.7.1 Skills, knowledge, training and experience

- To work safely, everyone on site needs to be competent or be closely supervised by someone who is.
- Anyone responsible for recruiting must ensure that the people they employ have, or are registered with an official process of obtaining, the necessary skills, knowledge, training and experience to gain competence in the job or task in hand (often demonstrated by ownership of an appropriate competency card).
- You will need to confirm and check the competence of contractors and persons coming onto site (such as those persons installing welfare facilities, erecting, altering or dismantling scaffolds, operating site plant, digging excavations and installing the supports or installing traffic management systems).

1.7.2 Consultation

- Employers must consult and engage with employees or their representatives in good time on issues that may affect their health, safety and welfare.
- Employees must be encouraged and be able to raise health, safety and welfare issues with their employer, directly or through their representatives.
- Where a trade union is recognised, ongoing issues requiring consultation should be facilitated through union appointed health and safety representatives. In non-unionised workplaces, consultation should be either directly with workers or through their elected representatives.
- As part of the site set-up, consideration must be given as to how effective employer/employee consultation will be achieved.

1.7.3 Co-operation

- CDM require all duty holders involved in a construction project to co-operate with each other in the interests of health and safety.
- How to achieve this co-operation is best decided at an early stage and, where possible, before the site is set up.



Regardless of your job and employer, on a construction site you have a legal and a moral responsibility to ensure you work safely and don't endanger yourself or others by your actions or omissions. You must communicate, co-operate and co-ordinate your actions with site management and other interested parties.

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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

Every year fires occur during construction, demolition, refurbishment and maintenance contracts. Fires are often small and are not reported to the client or enforcing authorities. Larger construction site fires or explosions tend to make the headlines and are reported.

Fires not only put workers' lives at risk but can also result in damage commercially and to reputation, and they invariably lead to lengthy delays in completion time. This chapter covers the requirements of duty holders to properly plan, manage and co-ordinate fire safety. It also gives advice on how fire risks can be identified and the control measures that should be implemented.

2.1 Introduction

Fire precautions should be implemented from the outset through design, procurement and completion of all construction projects. The risk of fire is an ever-present threat during all phases. As the work progresses, the nature of the fire risk is likely to change as different work starts and is completed. Therefore, not only must the risk of fire be managed continuously, but the management process must allow for the changing nature of the risk. Proper planning and adoption of safe working practices can prevent the majority of fires. Planning includes addressing fire risks at the design stage, and ensuring that the principal contractor, or contractor (if only one), enforces safe working practices. The client, principal designer and principal contractor must be satisfied that the risk of fire is, and will be, proactively managed.



It is essential that everyone on site knows what to do if the fire alarm sounds, whether it is for real, a practice drill or a false alarm.



The Fire Protection Association's *Fire prevention on construction sites* (Joint Code of Practice) is the code that applies to activities carried out before and during the procurement, construction and design process – not the completed structure – and should be read in conjunction with all current legislation and HSG168: *Fire safety in construction*.

If compliance with this code forms part of the insurance contract, non-compliance could possibly result in insurance ceasing to be available or being withdrawn, resulting in a possible breach of a construction contract that requires the provision of such insurance.

2.2 Important points

- Fire safety must be an integral part of a construction project, from planning and monitoring through to completion.
- Legislation requires a suitable and sufficient fire risk assessment to be carried out by a responsible person, or someone appointed to do so with sufficient experience and qualifications.
- The correct fire safety materials and resources should be budgeted for and procured.
- Everyone involved in the construction project should know who the responsible person is (they may change during the project).
- Fire risk assessments should comply with the Regulatory Reform (Fire Safety) Order 2005 (FSO).
- Fire risk assessments must address fire prevention and protection measures. They must be reviewed periodically due to the changing nature of the construction activities and introduction of new hazards, processes and personnel.
- Fire safety procedures should be developed and adhered to. These should be communicated to everyone on site so they know what action to take in the event of an emergency.
- A hot-work permit scheme should be implemented to control the way hot-work activities are managed and controlled.
- Plan where flammable materials (such as timber, gases, adhesives and flammable liquids) will be stored.
- Additional fire precautions may be required where on-site accommodation is provided.



Fire safety can be a complex issue on large, fast-moving sites with significant numbers of people to evacuate. Where the expertise is not present on site or within the company, it is advisable to seek the advice of a fire safety specialist.



In this chapter, in most cases, where the word *construction* appears the point being made will also be applicable to demolition. However, for simplicity's sake, only the word *construction* will be used.

2.3 Nature of fire

2.3.1 How a fire can start

Three factors need to be present to enable a fire to start.

Fuel or combustible material. These can be liquids, solids, gases or any materials that will burn and support the combustion, given sufficient amounts of heat and air or oxygen.

Heat or ignition source. Every fuel has an ignition temperature. All solids and liquids give off vapour when heated, and it is this vapour that ignites.

Air (or other supporter of combustion, such as oxygen) which is always there to sustain fire, providing that the other two factors are present.

Once a fire has started, combustion will continue as long as these three factors are present. If any one of these factors is removed, the fire will be extinguished. This can be achieved by:

- removing the fuel, leaving nothing to burn
- removing the heat by applying water to cool the burning material
- reducing or excluding the air by smothering the burning material. Foam, dry powder, CO₂ and fire blankets are all smothering agents.



The fire triangle

2.3.2 How fire can spread

Fire can spread in four ways.

1. Conduction.
2. Convection.
3. Radiation.
4. Direct burning.

Conduction is where heat is transmitted from one place to another along or through solid material (such as along a metal pipe or through a door or wall). The conduction of heat therefore has the potential to start a fire in a location that is remote from the original source of heat.

Convection occurs where superheated gases or heat arising from a fire ignite other combustible material or when particles of burning material in the circulating air are deposited in another place, causing another fire.

Radiation is the transfer of heat from the fire, through the air, to other flammable materials nearby, causing the temperature of those materials to be raised to their ignition point and then burn.

Direct burning is a combination of conduction, convection and radiation and is where the fire spreads and reaches other combustible materials and ignites them, adding further fuel to the fire.

2.4 Classes of fire

Fires can be placed into one of the following categories.

Class A. Carbonaceous material (such as paper, cloth, wood and rubber) often referred to as solid fuel fires.

Class B. Flammable liquids or liquefiable solids (such as oil, fat, paint and fuel). These can be subdivided into:

- B1.** Fires involving liquids that are soluble in water (such as methanol). They can be extinguished by CO₂, dry powder and water spray
- B2.** Fires involving liquids that are not soluble in water (such as petrol and oil). They can be extinguished by foam, CO₂ and dry powder.

Class C. Flammable gases or liquefied gases (such as propane, butane, hydrogen or acetylene).

Class D. Combustible metals (such as magnesium, sodium and phosphorus). They can be extinguished by specialist dry powder.

Class F. High temperature cooking oils or fats (such as those used in deep fat fryers in large catering establishments or restaurants).

Electrical fires (fires involving electrical apparatus or equipment) are not a specific fire class, as electricity is a source of ignition rather than a fuel.

However, they are still important to mention because they have their own special fire safety requirement.



For further information on classes of fire refer to Appendix A.

2.5 Legislative requirements

The main legal requirements that cover fire prevention and control on construction sites are the **Regulatory Reform (Fire Safety) Order 2005 (England and Wales)** (FSO), the **Fire Safety (Scotland) Regulations 2006**, the **Fire Safety Regulations (Northern Ireland) 2010** and the **Construction (Design and Management) Regulations 2015** (CDM).



For further guidance refer to the HSE's *Fire safety in construction* (HSG168).

2.5.1 Regulatory Reform (Fire Safety) Order 2005 and fire safety regulations

There are three main pieces of fire legislation in the UK that apply to all workplaces and premises.

- FSO (England and Wales).
- Fire Safety Regulations (Northern Ireland) 2010.
- Fire Safety (Scotland) Regulations 2006.

These apply to both the actual construction site and site offices, as well as to the head offices of a company that could be separate from the actual site address. The legislation covers fire prevention, detection, firefighting and emergency lighting routes during construction. The legislation requires that the project should have in place a valid **fire risk assessment**. The fire risk assessment should be reviewed periodically to take into account any changes in the construction activities and processes.

The legislation places duties on the **responsible person**: the employer or person who has control of any part of the premises. During the construction phase, the principal contractor or contractor will often be identified as the responsible person, and they must appoint a **competent person** who is sufficiently skilled and knowledgeable to help them fulfil the required duties. This person must be clearly identifiable during all stages of the construction project.

If the project involves refurbishment of occupied premises, the client may also have control or partial control. Therefore, both parties will need to agree and be sure of their responsibilities. The responsible person must make sure that the fire risk assessment is in place and remains up-to-date.

2.5.2 Construction (Design and Management) Regulations 2015

CDM (part 4) covers the following requirements on construction sites.

- The prevention of risk from fire, flooding or asphyxiation.
- Emergency escape routes and exits.
- Emergency procedures.
- Fire detection and firefighting.

There is a requirement under CDM for all duty holders to co-operate and co-ordinate their activities at all phases of the project.

Clients must make sure that arrangements are in place to manage health and safety from the start to completion of the project. This includes ensuring that a construction phase plan (including the **fire safety and evacuation plan**) is in place before work starts. During the construction phase the **principal contractor** or **contractor** (if only one) must make sure that the construction phase plan, along with the fire risk assessment and fire safety and evacuation plan, remain up-to-date.

The responsible person should be named in the fire safety and evacuation plan. This person may change throughout the construction phase. The named person must be aware of their responsibilities and any management actions or duties they are required to perform/**Principal designers** and **designers** need to identify and ensure that all potential fire hazards have either been eliminated or reduced so far as is reasonably practicable. They should consider the use of non-combustible and non-flammable materials to reduce the fire load.

If specifying a timber-frame design they must give consideration to the surrounding environment to assess the potential risk of fire spreading to neighbouring properties. They should also look to avoid processes that require hot works on site, therefore reducing the risk and amount of controls and management associated with hot-work permits. Additionally, they should work closely with site managers to ensure that compartmentation of the structure happens as early as possible, smoke dispersal arrangements and escape stairs are in place and to allow for the early installation and use of fire alarm and firefighting systems to protect against the rapid spread of fire during the construction phase.



Compartmentation

Compartmentation should be given priority by designers and contractors when considering fire precautions during construction. Effective compartmentation will stop fire spreading rapidly from one part of a building to another.

A building can be sub-divided by means of fire-resisting walls, floors and ceilings. Holes and openings must be protected by using fire-resisting materials.

Whilst this publication looks at the issues about fire risk during construction, it is worth noting that the final *Independent Review of Building Regulations and Fire Safety* places greater responsibility on duty holders for maintaining the 'golden thread' of information about the building structure and materials, detailing the maintenance, testing and inspection routine as well as how fire risk assessments have been undertaken and actions implemented. A comprehensive Fire and Emergency File and digital record will be transferred from the client to the dutyholder, giving them the golden thread of information needed to manage the building safely, and evidence to inform the safety case. This must be updated by the dutyholder so that it remains accurate throughout the lifecycle of the building.



For detailed information on the *Independent Review of Building Regulations and Fire Safety* visit the website.

2.5.3 The Building Safety Act 2022

The Building Safety Act is designed to improve the safety of buildings and minimum standards in response to the fire that took hold of Grenfell Tower in London in 2017. It makes changes that mean owners will manage their buildings better, and the construction industry has the clear, proportionate framework it needs to construct more high-quality housing. The Act sets out rules to protect the design and construction of higher-risk buildings by:

- regulating higher-risk buildings
- raising safety standards in all buildings
- helping professionals in design, construction, and building control, to improve their competence.

Building control professionals who will ensure that the safety and quality of the built environment is adequate will be regulated by the Building Safety Regulator (BSR) to raise standards and increase public trust.

2.5.3.1 Changes to building control

From 1 October 2023, the BSR automatically assumed the role of building control authority for all higher-risk buildings. Developers will no longer have the option to choose who they use as their building control body. This means that local authorities and approved inspectors will no longer be able to supervise higher-risk building works. The BSR will continue to work closely with local authorities and fire and rescue services to ensure that the correct specialists and regulators are involved when making decisions. Local authorities will remain the building control body for other, non-higher-risk, buildings.

2.5.3.2 Changes to the building control approval process

A new process for gaining building control approval for work on higher-risk buildings also came into force on 1 October 2023. The aim is that this process will allow the BSR to regularly assess whether duty holders are continuing to take account of building safety measures properly, and that they are appropriately meeting all regulations throughout the design and construction process. Changes to the Act and linked legislations continue to be made. Therefore it is important to ensure that work is being carried out to the latest regulations check the website for further up-to-date information.



For further guidance, refer to the Building Safety Act 2022.

2.6 Liaison with the fire and rescue services

During the design phase the principal designer or designer should contact the fire and rescue service (FRS) and discuss access requirements and arrangements. At the construction phase the principal contractor or contractor should make contact with the FRS and provide an initial site plan. This plan should be updated and made available on site. The site layout plan should include the following.

- The fire and rescue access point(s).
- Emergency escape routes and stairs.
- Firefighting shafts or cores, lifts and hoist facilities.
- Location of fire call points and other firefighting equipment.
- Location of hazardous material (such as flammable and highly flammable material storage areas).
- Location of the assembly points.
- Floor loading limitations.
- Sprinkler installations.
- Positions of hydrants and any wet or dry risers.
- Details of any temporary buildings.



Specialist firefighting access equipment will be required where construction work will take place above 18 m. This must be brought to the attention of the FRS.

2.7 Responsible person

As mentioned earlier (refer to 2.5.1) a nominated responsible person is required under legislation. Often during construction this role will fall to the principal contractor, where appointed, or the contractor.



The Regulatory Reform (Fire Safety) Order 2005 – Responsible person

- This will be the employer, in relation to a workplace, if the workplace is to any extent under their control.
- In relation to any premises not falling within the bullet above:
 - the person who has control of the premises (as occupier or otherwise) in connection with the operating by them of a trade, business or other undertaking (for profit or not), or
 - the owner, where the person in control of the premises does not have control in connection with the operating by that person of a trade, business or other undertaking.



Refer to the Fire Safety Act 2021, which amends the FSO.

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The responsible person must be competent and must ensure the following:

- Carry out a suitable and sufficient fire risk assessment of the site, record the findings in full, and ensure that the assessment is reviewed and remains relevant.
- Tell workers about the findings of the risk assessment.
- Take general fire precautions to ensure, so far as is reasonably practicable, the safety of employees and relevant persons. These can be persons who are lawfully on the construction site or in the immediate vicinity of the site.
- Put in place and maintain fire safety arrangements. These arrangements must be appropriate to the size of the construction site and the nature of the activities. These arrangements should include safe storage and transport of flammable materials, maintenance of fire safety equipment, hot-work permits, monitoring and inspection arrangements.
- Plan for an emergency. This detail can be written into the fire safety and evacuation plan and should include adequate emergency escape routes and exits, firefighting and fire detection arrangements and liaison with the emergency services.
- Provide workers with fire safety information, instruction and training. This can include displaying fire evacuation procedures, information on the arrangements to take in the event of discovering a fire, fire evacuation drill practices and fire extinguisher training.
- Co-operate and co-ordinate where two or more responsible persons have duties at the same premises.
- That they do not appoint a person to assist them in making or reviewing a fire risk assessment unless that person is competent.
- Record the identity of any person appointed to assist them.
- Record the fire safety arrangements.
- Take reasonable steps to ascertain whether there are any other responsible persons who share or have duties in respect of the premises.
- Record the other responsible person's:
 - name and address in the United Kingdom at which they, or someone acting on their behalf, will accept notices and other documents
 - area of responsibility, i.e. the part of the premises for which they consider themselves to be responsible.
- If someone takes over as the responsible person, the previous one must hand over any relevant fire safety information to maintain records.

A fire safety co-ordinator and fire warden(s) can be appointed to help the responsible person to perform their role.



The responsible person could invite their local fire and rescue service to attend site and carry out a site inspection and familiarisation tour.

2.7.1 Fire safety co-ordinators and fire wardens

On larger or higher-fire-risk sites, the responsible person can appoint others to assist with the management of fire safety on site, and provide an increased level of management oversight in relation to fire safety. These can be either fire safety co-ordinators or fire wardens (and deputies). The appointment of deputies would be regarded as good practice on a complex site. The number of wardens will depend on the layout, site complexities and number of workers on site (a fire warden may also be known as a fire marshal).

On site, fire wardens or the fire safety co-ordinator (if appointed) would have the following responsibilities.

- Assist in implementing the fire safety and evacuation plan.
- Conduct weekly inspections of escape routes, firefighting equipment, access for emergency services and testing of alarms at detection devices on site.
- Keep records of all checks, inspections, tests and fire drills.
- Ensure that the hot-work permit systems are being followed and managed correctly.
- Liaise with security guards (where appointed) and fire and rescue services.
- Assist in fire evacuations.
- Ensure that offices and other accommodation are completely evacuated in the event of the fire alarm sounding (including practice drills) and conduct a roll call at the assembly point.

On large sites it is good practice for each sub-contractor to appoint a fire warden to account for their company's staff in the event of an evacuation. The location and number of staff can be identified by the use of an in/out board, signing in/out book or swipe card system that can be taken to, or accessed at, the assembly point and used for the roll call. Arrangements must be made for visitors to be logged into and out of offices so that, in the case of an emergency, they can be located quickly and escorted to a safe place.

Fire wardens and fire safety co-ordinators should be given sufficient time to perform their fire safety duties. They should receive training, specific to the role, which should be refreshed on a regular basis.

2.8 Fire risk assessment

There is a legal requirement for the responsible person to ensure that a site-specific fire risk assessment is carried out. They can either do this themselves or nominate a competent person to do it on their behalf. The responsible person on site will normally be the principal contractor or contractor. Fire-specific legislation and CDM apply here, to ensure that risks are identified and managed. The risk assessment should identify the activities carried out on site and the likelihood that a fire could start and harm people in and around site. If an enforcing authority (such as the HSE or the fire and rescue service) visit site, they will expect to see a copy of the risk assessment.



If you do not properly assess the risks, there is no way that you can adequately control them.

The principles of carrying out a fire risk assessment are broadly the same as for any other risk assessment.

There are five steps to carrying out a fire risk assessment.

- 1. Identify the hazards.** Consider how a fire could start and what could burn, and the types of construction material used and stored on site, as well as the location of the build and how close it is to other buildings. This process should start at the design stage.
- 2. Identify the people at risk.** Consider employees, contractors, visitors, immediate neighbours and people who may not be on site. Extra arrangements may be required for people with disabilities.
- 3. Evaluate the risks and plan** how to remove or reduce them and how to protect people from any risks that remain (residual risk).
- 4. Record, plan, inform, instruct and train.** Keep a record of the risks and actions taken, develop a fire safety and evacuation plan and ensure that people understand what they need to do in the event of a fire.
- 5. Review** the fire risk assessment regularly to ensure it remains valid and takes account of any changes on site.

On smaller projects the initial assessment might be valid all the way through to completion. On larger or complex projects, where there are a greater number of people to evacuate or there are long travel distances to a place of safety, someone with a detailed knowledge of fire risk management (such as a consultant) may be required.

In this scenario, it is likely that conditions around site will change as the project progresses meaning that periodic reviews and updates of the assessment will be required. Irrespective of the size or complexity of the project, anyone carrying out a fire risk assessment must be competent to accurately identify the hazards, assess the risks and put effective control measures in place.



For further information on the process of fire risk assessment refer to Appendix B.



A guide to choosing a competent fire risk assessor has been published by the Fire Risk Assessment Competency Council. For this and further documents to help you ensure you are competent to undertake risk assessments, visit the Fire Industry Association and Institution of Fire Engineers websites.

2.9 Fire safety and evacuation plan

The fire safety and evacuation plan should be based on the findings of the fire risk assessment. Like the fire risk assessment, the plan should be reviewed periodically so that it remains up-to-date and reflects the changing nature and hazards of the site.

A **fire safety co-ordinator** could be appointed to take responsibility for the fire safety and evacuation plan and its implementation.

The person who completes the fire safety and evacuation plan could be a member of the existing site management team or, where particular expertise is required, from a consultancy or specialist fire safety organisation.

The following are examples of sites where external expertise might be required.

- Those large in area, with a high number of people to evacuate via multiple escape routes.
- Where the structure being built or refurbished has several floors with limited access between them.
- Sites and structures with long or complex escape routes, with changes in direction and levels.

As a minimum the plan should include details of the following.

- The organisation and named individuals who have responsibility for fire safety.
- Arrangements for recording fire safety training given to site workers.
- Details of general site fire precautions, for example:
 - fire detection and alarm systems
 - temporary emergency lighting
 - fire extinguishers and fire points
 - fire escape arrangements and communications
 - procedures for calling the fire and rescue service.
- Fire service access facilities and co-ordination, including the provision of operational rising mains.
- Fire drills and training.
- Hot-work permit procedures, where hot works cannot be avoided.
- Site accommodation, including location, fire protection, construction and maintenance.
- Location and control of smoking arrangements.

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- Security measures to minimise the risk of arson on site.
- Material storage arrangements and waste control regime, particularly where flammable and highly flammable materials will be stored on site.
- Maintenance of temporary electrical installations.
- Details of any coverings that will be used (such as monoflex sheeting), and a record of checks to make sure that they are fire retardant.
- Arrangements for plant and vehicles, in particular storage and refuelling arrangements.
- Measures to prevent the spread of fire on site and to any adjoining properties.

2.10 Common sources of fire risk

2.10.1 Hot work

Failure to adequately control hot work has been the cause of many construction site fires, many of them on refurbishment and maintenance jobs due to cutting, burning and welding operations and intense heat from halogen lights.

The Fire Protection Association (FPA) introduced a hot work passport scheme to enhance the level of competence of those involved in work with the potential to produce ignition sources. The passport is valid for five years and is suitable for employees involved in carrying out hot works or with responsibility for supervising or managing hot works.



Hot work

An activity or process that generates flame, heat or spark and introduces a foreseeable risk of fire or explosion through a source of ignition created by tools or equipment, such as the examples shown below.

- Intentional source of ignition arising from working methods with or without the use of a naked flame (for example, welding, flame-cutting, soldering, blazing, heat-stripping of paint and heat-sealing of roof membrane).
- Unintentional generation of heat or sparks, by the use of power or hand tools (for example, grinding and the use of disc cutters).

A hot-work permit system must be introduced to make sure that such work can be carried out in a safe and controlled environment.

It is essential that anyone involved in hot works is:

- aware of the fire risk in the particular location where work is to be carried out
- aware of how heat might be transmitted to other areas
- trained and competent to use the equipment that will produce the source of heat
- trained and competent to select and use portable fire extinguishers.

Precautions must be taken where heat from a work process could be transmitted to other combustible materials (for example, where hot works are carried out on steelwork or pipes adjacent to, or passing through, flammable materials, such as studwork walls or timber floors). Where this kind of operation is anticipated, it is essential that all combustible materials, including liquids, are protected before any work is allowed to start. Special care should be taken when cutting or welding is carried out above flammable materials. Precautions must be taken to prevent sparks or hot fragments of metal from dropping onto and igniting the flammable material below (such as removing it or covering it with fire-resisting material).

Hot work should cease at least **one hour**, or **two hours** on high-risk sites (such as large, timber-frame projects), before the site closes, with the area actively monitored for signs of fire until the site is vacated. Checks must include an inspection of cavities, around eaves, behind studding and into other voids where smouldering material would otherwise go undetected.

The introduction of a permit system should be considered for all hot work, except where the risk of fire is considered to be low. Hot-work permits are formal management documents that should only be issued by someone with official authority to do so. They should normally include the following.

- The location where the work will be carried out
- A description of the hot work to be carried out
- The proposed start time and duration of the work activity
- The time at which the permit expires (the work must be completed or cease at that time).

Other precautions to be taken, which should be reflected in the hot-work permit, include instructions for:

- clearing all combustible material from the surrounding area
- checking for combustible material in adjacent hidden locations
- having at least two suitable and serviceable fire extinguishers at the place of the work

- maintaining a careful watch for the early signs of fire during the work
- protecting combustible material that cannot be cleared (for example, by the use of non-combustible matting)
- on completion of the hot works, carrying out checks and monitoring the area until the site closes.

Even where a formal hot-work permit system is not in operation, because of a perceived low risk of fire, basic fire safety precautions (such as having a serviceable fire extinguisher to hand) are still required. The following precautions must be observed when using **bitumen boilers**.

- The distance between the boiler and liquid petroleum gas (LPG) cylinder must be at least 3 m, unless a suitable protective shield is fitted.
- Do not place boilers inside a building or in an enclosed space.
- Stand the boiler in a tray, with a capacity greater than the boiler, to capture any spills or leaks.
- Additional LPG bottles must be stored at least 6 m away from the boiler.
- The boiler must never be left unattended, moved or towed whilst the burner is alight.
- For extra safety only use boilers that are thermostatically controlled.
- If possible avoid taking boilers onto roofs; if this cannot be avoided, place the boiler on a non-combustible, insulating base.
- Appropriate firefighting equipment must be located nearby.



Never use water to fight a bitumen fire.



For detailed information refer to the HSE document *Health and safety in roof work* (HSG33).

2.10.2 Flammable liquids and gases



You must be careful when LPG is being used on your site; the person doing the job must be competent and you must ensure that they are.

LPG should be stored in purpose-designed, fireproof compounds or cages that prevent the accumulation of vapour in the event of a leak. Gas cylinders are capable of withstanding normal UK weather conditions, including temperatures between -20 °C to +55 °C. As such, they may not require protection from the weather when in storage. The floor should be constructed of concrete or other rigid, permanent, non-combustible and non-porous material. The floor should not be constructed of bricks, slabs, flags, asphalt or tarmacadam. Signs must be clearly displayed indicating the presence of LPG, and prohibiting smoking and the use of any naked flame in the area of the store.

Precautions for the use of LPG include the following points.

- Minimise the amount of LPG on site at any one time.
- Turn off cylinder valves before connecting or disconnecting any equipment.
- Check cylinders and all associated equipment before use; do not use if there are signs of damage or leaks.
- Small, easily portable LPG cylinders and canisters should ideally be removed from site after work.
- Use leak-finding fluid or soapy water to try to detect the exact location of leaks; the smell should provide the initial indication.
- Store LPG cylinders:
 - with their valves uppermost, apart from cylinders used to provide fuel for LPG-powered plant, which are used and stored on their side
 - away from oxygen cylinders, highly flammable liquids, oxidisers, toxic or corrosive gases or substances
 - away from untrapped drains and cellars
 - a distance of at least 3 m from other substances, although they may be kept in the same compound.
- **Never** store LPG cylinders below ground in cellars or basements.



LPG cylinders must never be stored in unventilated metal boxes or site accommodation.

Ignition of released LPG, where the concentration exceeds 2%, can result in fire or, if confined, an explosion. If a leak does not immediately ignite, and the LPG and air mixture drifts from the point of release, it will gradually become more diluted. However, if the concentration still exceeds 2%, ignition could cause a flash or cloud-fire back to the point of release. A leak of LPG may be noticed either by the smell or the noise of the gas escaping. Leaks must not be traced with a lighted match or naked flame as this would almost certainly cause an explosion. Only soapy water or a proprietary leak-finding fluid should be used. If it is suspected that LPG has leaked inside a building, no attempt should be made to touch any electrical apparatus.



Never turn on light switches or any other electrical equipment if a leak of LPG is suspected.

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Open all doors and windows, if it is safe to do so, and leave immediately. Do not re-enter the building until advice has been sought and you are told it is safe. Where LPG is used for fixed heaters, cookers and lighting within welfare facilities, the cylinders and regulators must be kept outside and the gas supplied using rigid copper piping.

The use of flexible hosing is permitted only between the cylinders and changeover valves or manifolds, and for the final connection to appliances, but this must be kept as short as possible. There should be ventilation grills at low and high level to maintain ventilation. Check that these have not been blocked or covered to prevent cold drafts and install carbon monoxide alarms.

All pipework should be easily accessible for inspection, but protected against accidental damage. Any work on LPG pipework or other parts of a fixed installation, including testing, must only be carried out by appropriately trained (Gas Safe registered) persons.

Inspections of all appliances must be carried out before use. If soot forms or smells occur, place the appliance in quarantine. Find out the reason for the problem and have it corrected.

Before using LPG equipment in an enclosed space a risk assessment must be carried out. It is essential to ensure that there is adequate ventilation, which may have to be forced. This is necessary to ensure full combustion and also to make certain that the products of combustion, other fumes and excess oxygen from any cutting apparatus are removed. Enhanced fire safety precautions and the use of atmospheric monitoring should be considered.

Wherever practicable, cylinders used with operations in confined spaces should be located in a safe area, preferably in the open air. The supply pressure should be reduced to the lowest practicable level on leaving the source of supply.

When working below ground, the number of cylinders must be kept to a minimum. All cylinders and hoses should be removed as soon as work has finished or if it is interrupted for a substantial period (for example, overnight). LPG cylinders must not be taken into confined spaces, as defined in the Confined Spaces Regulations, unless exceptional safety precautions are taken.

Highly flammable and explosive substances (such as adhesives, solvents and paint) should be stored as follows.

- Where site conditions allow, in an external, secure, purpose-built compound.
- Where external storage is not practical, in a suitable secure, internal, fireproof storeroom with fire detection and suppression systems installed.
- In a metal, lockable cabinet or bin, for immediate or imminent use at the place of work.

Where it is necessary to store bulk quantities of highly flammable substances, the requirements of the storage area are shown below.

- Located outdoors in a position away from watercourses and drains and protected against vandalism or impact from vehicles or plant.
- Be stored within an impervious bund with a capacity greater than 110% of the total potential stored contents. For multi-tank storage the capacity must be 110% of the largest tank or 25% of the total combined tank capacity, whichever is greater.
- Protected against direct sunlight.
- At least 2 m away from nearby buildings or boundaries, except where the boundary of the store forms part of a solid wall; cans or drums may be stacked up against that wall up to 1 m from the top.
- Fitted with appropriate signage to indicate the content (such as 'highly flammable' or 'flashpoint below 32°C').
- Fitted with an 'EX' sign when an assessment under the Dangerous Substances and Explosive Atmospheres Regulations (DSEAR) indicates that an explosive atmosphere could be present in the event of a leak.
- Fitted with 'No smoking' or 'No naked flames' signs.
- Kept locked when access is not required.
- Equipped with intrinsically safe electrical circuits and fittings where electrical supplies are necessary.
- Equipped with dry powder or foam fire extinguishers.
- Equipped with a quantity of absorbent material, to soak up any spilt liquids, and a suitable container for the collection and safe disposal of the contaminated absorbent.
- If necessary, equipped with racking made from a non-ferrous metal or other non-combustible material.



Image showing bad practice. Designated refuelling and cutting points should be used or, better still, another type of powered equipment should be used when working in timber-frame or enclosed buildings



EX icon hazard sign indicating an explosive atmosphere

! The secure storage of dangerous materials in a proper compound or cabinet is as equally important as using them in a safe manner.

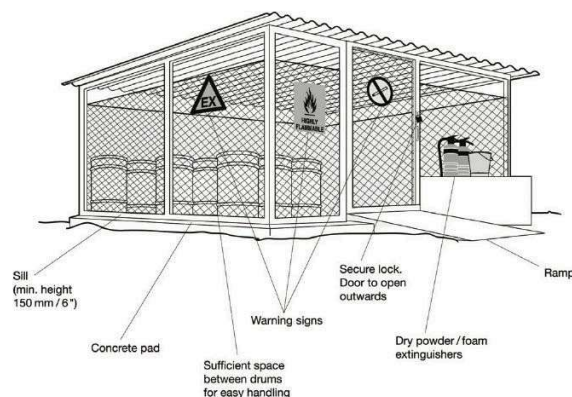
If for any reason it is not practical to store highly flammable or other dangerous substances outside (for example, if the footprint of the structure under construction completely fills the land available), the conditions above for outdoor storage apply, and the DSEAR assessment must consider whether risks of storage indoors are acceptable. If it concludes that it is not safe, off-site storage and transportation must be arranged.

In deciding whether or not the risks associated with indoor storage are acceptable, it is likely that expert advice will be required to determine the suitability of the proposed storage area with regard to its fire and explosion resistance.

It is recommended that the maximum quantities that may be stored in cabinets and bins are no more than 50 litres for highly flammable liquids (and flammable liquids with a flashpoint below the maximum ambient temperature of the workroom/working area) and no more than 250 litres for other flammable liquids with a higher flashpoint of up to 55°C.

Under no circumstances should storage areas be located underground as this would prevent the dispersal of leaks from gases and vapours that are heavier than air.

All signs must conform to the Health and Safety (Safety Signs and Signals) Regulations 1996.



A secure dangerous substances storage area



When storing highly flammable materials at the workplace, ready for use, only store one day's supply or less.

On many sites there will be a need to store bulk quantities of vehicle fuels, particularly diesel oil, usually in a bowser or drum. If storage drums are used, the same conditions apply as for the storage of other highly flammable and explosive substances.



Carelessly dispensing fuel into vehicles or plant when the engine is still hot has been the cause of many serious fires and injuries.

Where fuel oil is to be dispensed into containers or vehicles, the outlet must be fitted with a valve or nozzle, which closes automatically when not in use and cannot be fixed in the open position unless an automatic cut-off nozzle is used on the end of the hose.

The hose and nozzle should be housed in an enclosed, secure cabinet with a drip tray when not in use, or have a lockable valve where the feed leaves the tank. Hoses should be kept locked or kept within the bund when not in use.

Any permanent valve or tap that opens directly from the tank must:

- be located within the secondary containment system (bund wall)
- discharge vertically downwards
- be locked off when not in use.

Finally, no vent must discharge outside of the bund in the event of an overfill.

Generally, petrol is only used in relatively small quantities for generators and some hand tools (such as chainsaws) but, however small the quantity, it must be stored safely.

Petrol will usually be stored in 20 litre metal jerry cans or five-litre, purpose-made plastic containers. The jerry cans or containers must be kept closed, except when the content is being decanted.

The positioning of fire extinguisher points, and the types of extinguisher, must be based in part upon where flammable construction materials will be stored and used.

Storage areas for flammable substances and gas bottles should be located away from emergency escape routes and assembly points.

For more detailed information on flammable liquids refer to the following HSE guidance.



- **Safe use and handling of flammable liquids (HSG140).**
- **Storage of flammable liquids in tanks (HSG176).**

2.10.3 Material storage

Many materials (such as timber) and substances used in the construction process will readily catch fire if exposed to a source of ignition.

Where practical, all combustible materials should be stored outside – not within a building. Ideally, all materials will be stored in a locked compound or storage container, with adequate separation between different types of product. Where internal storage is inevitable, the storage area must:

- be constructed and arranged so that the spread of fire would be limited
- be separated from the rest of the structure by a partition that provides a **minimum of 30 minutes' fire resistance** and doors must be fire resistant and self-closing
- be located away from evacuation routes so that it does not pose a threat to people trying to evacuate the site.

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2.10.4 Storage of waste

Most construction sites generate large quantities of waste that present a potential fire risk. Good housekeeping should be maintained and waste should be cleared from site on a regular basis. Poor housekeeping will increase the fire loading and has the potential to block access to fire alarms, firefighting equipment and escape routes, putting people at risk.

Examples of poor planning and housekeeping that increase fire risk are shown below.

- Work being carried out in designated escape routes (for example, tower scaffolds erected on staircases).
- Flammable liquids and gases not being returned to their designated storage areas after use.
- Access routes for the emergency services being blocked, preventing access to all parts of the site.



A buildup of combustible material causing a fire risk

Skips and other waste disposal containers should be placed **a minimum of 3 m away** from the structure under construction or the means of escape from it. If this cannot be done, the following actions should be taken.

- Encourage the regular clearing and segregation of waste.
- Prevent build up of a significant fire load.
- Make sure skips are easily accessible to vehicles that are facilitating removal or replacement of the skips.
- Place skips away from canopies and overhanging eaves.
- Place skips against a fire-resisting wall, which is high enough to prevent fire from reaching other flammable parts of the structure.
- Ensure that personnel working on site use only a chute made of non-combustible materials.
- Restrict the amount of flammable material placed within waste disposal containers.

The following points should also be considered.

- Site rules should state that contractors must keep their areas clear of waste at intervals appropriate to the nature and volume of the waste created, but at least daily.
- Skips should be emptied or replaced before the contents represent a fire risk.

Flammable waste materials should not be allowed to accumulate but instead be cleared away promptly to hazardous waste skips located outside the structure; the more waste material there is within the structure, the greater the fire loading.

2.10.5 Electrical installations

Electrical installations should be professionally designed, installed, inspected and maintained, in accordance with the **Electricity at Work Regulations 1989**. This will ensure that the installation has the required capacity to cope with the intended loads and will remain in a safe condition. A regime of regular inspection and maintenance of the system will be required, with any necessary alterations being designed and installed by competent persons.



Unauthorised alterations must not be made to the distribution system.

Halogen light fittings can get hot and cause materials near them to ignite. Historically, they have been a common source of ignition for site fires and it is good practice to only use them where other forms of task lighting are not suitable and the fire risk assessment shows that it is safe to do so (some sites have banned the use of halogen lights). The failure of electrical hand tools and mains-powered office equipment has also been the cause of fires. The regular testing of portable electrical appliances and hand tools (PAT – portable appliance testing) will confirm whether the internal insulation is breaking down or there are other defects rendering the equipment a potential fire hazard.

PAT testing helps an employer maintain regular maintenance checks on portable electrical appliances and equipment, and evaluate the regularity of those tests according to the risk. Portable electrical equipment should be switched off and disconnected from the supply when not in use for extended periods. This extends to equipment that might be left on overnight (such as computers or photocopiers).

Possible causes of electrical fires are shown below and on the next page.

- Overheating of poorly maintained or damaged equipment and hand tools.
- Overloading of electrical circuits.
- Overloading of electrical sockets in site accommodation.
- Accumulation of combustible rubbish alongside cables (frequently in ceiling voids) or distribution boards.
- Bypassing of circuit safety devices, rendering them inoperative.

- Fitting of the incorrect rating of fuse, circuit breaker or residual current device (RCD).
- Unauthorised modification of the site electrical distribution system.
- Failure to carry out the periodic testing and maintenance of electrical circuits and safety devices.
- Sparking within electrical equipment or tools igniting explosive airborne particles (dust) or fumes.
- Accidental damage to equipment and conductors that have not been adequately protected.

For further information refer to the following HSE publications.



- ***Maintaining portable electrical equipment (HSG107).***
- ***Maintaining portable electric equipment in low-risk environments (INDG236).***

2.10.6 Bonfires

Environmental factors, risk of uncontrolled fire and/or Local Authority restrictions make the controlled burning of rubbish on site an unacceptable practice. The potential for the reuse or recycling of waste materials should always be considered with alternative arrangements made for the proper disposal of rubbish and waste.

2.10.7 Plant and equipment operations

Most engine-driven plant and equipment will generate heat during use. Excessive heat can be generated if the plant or equipment is poorly maintained or if it is being used beyond its design capacity. Furthermore, plant represents a fire hazard when it is being refuelled or stored away, under cover, whilst still hot. The lead-acid batteries used in plant and other vehicles give off an explosive gas when the batteries are being recharged, if the batteries are of a type where the filler caps have to be removed or loosened. The same will apply to any type of electrical equipment (for example, temporary traffic light units, which incorporate this type of battery).

2.10.8 Protection of finished surfaces

Once a building is nearing completion, it is common practice to use temporary coverings to protect finished surfaces and installed equipment during the remaining fitting-out phase. These coverings often take the form of plastic sheeting, fibreboard or similar materials, some of which may be flammable. They can consequently assist in spreading fire. Ideally, purpose-made, fire-retardant materials should be used.



All temporary flexible covering materials used for the protection of internal surfaces or fittings incorporated into the building or structure should be compliant to *Fire requirements for protective covering materials (LPS 1207 (internal) and LPS 1215 (external))*.

If non-fire-retardant materials are used, the protective covering should be fitted as late as possible in the project and the fire-protection measures must be suitable to cover the enhanced risk.

2.10.9 Scaffolding sheeting

If scaffolding is to be sheeted and used as a means of escape, the sheeting must be made to a flame-retardant standard (LPS 1215). Consideration should be given to ventilation (to clear smoke) and access for the emergency services.

2.10.10 Arson

Arson has been the suspected cause of many construction fires. It should be identified on the risk assessment as a potential hazard, and the control measures should be written down and included in the site fire safety and evacuation plan. Flammable materials should not be stored where they can be seen and set alight from outside the site boundary. Therefore, hoarding and fencing around the site perimeter, and securing access points, can help prevent arson or the temptation of arson.

Security measures (such as 24-hour security, CCTV cameras and intruder alarms) can help to secure the site from arson risks. If security guards are to be used, part of their patrol can include fire checks during the day, night, holidays and weekends. Some other ways to prevent arson are listed below.

- Only allow authorised people on site.
- Employ site security to prevent unauthorised access to the site out of normal working site hours.
- If necessary, remove flammable liquids, LPG and other gases and combustibles whilst the site is closed.
- If the completed project includes security fencing, this should be brought forwards and used during the construction phase.
- Provide security lighting.
- Log any suspected incidents so that preventative action can be taken.



If employing security personnel the duties they undertake must be covered by a risk assessment.

2.11 Higher-risk projects

Projects that will require extra precautions could be timber-frame structures, multistorey buildings or high rise construction sites, or those where significant risks have been identified during the risk assessment. Some of these are explained in more detail below.

2.11.1 Timber-frame structures

Large, timber-frame buildings under construction without the appropriate protection measures and controls in place pose a significant risk to construction workers, fire-fighters, adjacent buildings and members of the public.

Timber-frame structures are at their most vulnerable when they are being erected and before any cladding or protective skins are installed. Each structure should be compartmentalised and clad at the earliest practical opportunity. Designers should give consideration to the use of materials that have been given a fire protective or retardant treatment.

Compartmentalising the structure at the earliest opportunity prevents the rapid spread of fire and supports safe travel distances for anyone who has to evacuate the structure in an emergency.



High-risk timber-frame buildings require additional fire-prevention measures.

Consideration must be given to using fire-resistant boarding to temporarily fire-stop ducts and shafts until the permanent fire-engineered solution is installed. Temporary or permanent fire-resisting doors, panels and fire stopping should be installed as early in the construction programme as is practical.

Temporary accommodation units, site cabins and offices should be situated at least 20 m away from the timber-frame structure to act as a fire break if a potential fire did break out.

Where there are multiple, timber-frame structures on a site, the fire risk assessment must consider how a fire in one of them could be prevented or slowed from spreading to the others or to adjacent premises. To this end, temporary non-combustible cladding could be used to cover exposed timber and other combustible construction materials prior to the final cladding being installed.



High-risk timber-frame buildings require additional fire-prevention measures

The Structural Timber Association has published several guides (freely downloadable from its website) related to fire safety in timber-frame construction, as part of its safe site strategy.



- **Fire risk induction pack for site workers.**
- **Design guide to separating distances during construction.**
- **Risk assessment checklist.**
- **Guidance notes on wireless fire alarms.**
- **16 steps to fire safety.**

2.11.2 Multistorey buildings

It is advisable that the fire and rescue service are informed that any construction work or renovations are taking place. The Building Regulations 2010, for example, require additional measures on buildings above 18 m (or seven storeys or more). The important factor, in relation to construction site fire safety, is that the risks associated with the build can be adequately controlled, and that the GFPs needed in the event of a fire are sufficient to ensure that every person on site can return to the ground and to a place of safety.

More-complex and very high-rise projects are beyond the scope of this guidance. You may need additional help with these from a competent specialist with the necessary skills, knowledge and experience of both fire safety and construction of high-rise buildings.

This type of structure, during construction or refurbishment, can pose particular problems with regard to the management of fire risk. Incomplete or absent fire-engineered solutions can result in the rapid spread of fire, leaving workers on the upper floors particularly at risk. Furthermore, it is possible that in some circumstances it will not be practical to supply sufficient water for firefighting purposes to the higher levels. In these circumstances a specific risk assessment must be undertaken to establish how the safety of those working on the upper levels can be assured in the event of a fire.

Designers have a major impact on fire safety, not only in their selection of materials and methods of work, but also in ensuring the following.

- Temporary, fire-resistant compartmentalisation or other fire-engineering solutions are incorporated until permanent fire stopping is in place.
- With new-build or refurbished premises, a working fire alarm system is extended as each level of the building is completed so that it can be heard in all parts of the building at all stages of construction.
- The fire alarm system is electrically operated (hard-wired or wireless) with break-glass call points and sounders at each level.

- The rising water main(s) is installed and commissioned at an early stage before interior work starts.
- If some floors are already occupied, the risk assessment and fire safety and evacuation plans allow for:
 - the integration of the fire alarm systems for both the construction areas and the occupied floors
 - any restrictions that prevent construction workers accessing the occupied part of the building, particularly escape routes
 - the possibility that construction workers may not be allowed access to occupied parts of the building.
- Access to all staircases, or other protected routes, is maintained at all times, as far as possible.
- One staircase is designated as a firefighting staircase for the sole use of the fire and rescue service during an emergency.
- Fire doors with self-closers are fitted to protect the escape stairways and provide adequate fire stopping.

2.12 Fire detection and emergency planning

As detailed in Regulation 32 of CDM 'suitable and sufficient fire-fighting equipment and fire detection and alarm systems must be provided and located at suitable places'. This requirement includes examination, testing, maintenance and signage. Additionally it stipulates that 'each person at work on a construction site must, so far as is reasonably practicable, be instructed in the correct use of fire-fighting equipment that may be necessary for the person to use'.

The responsible person must consider the following aspects, 2.12.1 to 2.12.6 below, at all stages of the project and review and revise as necessary. Changes identified should be reflected in the fire risk assessment and fire safety and evacuation plan and communicated to those who are at risk.

! The emergency plan must be reviewed and updated on a regular basis as the structure develops and site conditions change.

2.12.1 Fire alarm systems

In the event of a fire it is essential that the alarm is raised as quickly as possible so that everyone on site can evacuate and reach a place of safety. As the site changes, the fire alarm system may need to be modified so that coverage across site is maintained.

- Alarms must be sufficient in number and located so that everyone on site is aware that the alarm has sounded.
- Manually operated sounders (such as hand bells, whistles and klaxons) will only be practicable for use on sites where they can be heard above any site noise and they must be readily available.
- Temporary hard-wired systems operated from call points are best when the site is large and covers multiple floors.
- If using a wireless fire alarm system, consideration must be given to ensure that the signal remains constant and uninterrupted. The Fire Protection Association's Joint Code of Practice states components of an automatic fire detection and alarm system should be marked as complying with BS EN 54.
- Flashing lights or vibrating pagers could be used if it has been identified that the alarms cannot be heard or where there is continuous loud noise.



Sites must have an effective fire alarm system

The operation and effectiveness of the fire alarm system should be routinely checked and serviced. Records of these checks should be maintained. If the project is a high-risk site or includes temporary accommodation units then automatic fire detection systems should be linked to the sounders. Domestic-type smoke alarms are not considered suitable on sites. However, they may be suitable on low-risk sites or in a small, temporary site office.

2.12.2 Fire detection systems

During the early stages of new build projects it is inevitable that the only way of detecting fire is visually; someone will see a fire and raise the alarm. However, on larger and high-risk sites and as the areas become enclosed, with increasing amounts of flammable materials present, the benefits of an automatic fire detection system become obvious.

Where a permanent automatic fire detection system is planned, phased activation of the system should be programmed in to follow the progress of the construction as early as possible.

During refurbishment work, if an automatic fire detection system already exists it should remain live throughout the work, where it is practicable to do so.

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2.12.3 Fire action notices

Fire action notices should be displayed in prominent locations across site and should include the following.

- Actions to take in the event of a fire.
- The location of the nearest fire assembly point.
- Instructions for calling the fire and rescue service.

Certain people on site may have additional duties in the event of a fire. For example, security guards may be responsible for providing unobstructed access for emergency vehicles, and supervisors may be responsible for undertaking a head count of their workers to ensure everyone is accounted for at the assembly point.

Principal contractors or contractors must ensure they have briefed everyone on site about the site emergency procedures. This is normally covered during the induction. However, they may need to provide additional briefings (such as a toolbox talk) as escape routes change.

2.12.4 Fire drills

Emergency procedures should be tested by carrying out regular fire drills, at least once every six months. Details of these tests should be recorded (for example, in a designated fire log book).

Details of the test could include a note on how long it took everyone to evacuate and any findings (for example, if workers reported they couldn't hear the alarm or if fire escape routes were blocked or unavailable) so that corrective action can be taken.

Fire drills will also identify if there are any bottlenecks, so alternative routes can be identified before a real emergency occurs. No-notice fire drills provide the most useful information on the effectiveness of the evacuation procedure. However, it is recognised that these may not always be practicable.

2.12.5 Means of escape

Escape routes must be located so all parts of the site have access to a place of safety, in the open air and at ground level. There should be at least two means of escape from the structure at all times, including from the basement and from roof level. Travel distances must be appropriate for the level of fire hazard.

Where dead ends exist they should be closely monitored and travel distances must be kept to a minimum in these situations. Any openings along a dead end route must be able to restrain a fire for half an hour.

Maximum travel distances from the fire to the safe area			
	Low level of fire hazard	Medium level of fire hazard	High level of fire hazard
Enclosed structure			
Alternative	60 m	45 m	25 m
Dead end	18 m	18 m	12 m
Semi-open structure			
	Low level of fire hazard	Medium level of fire hazard	High level of fire hazard
Alternative	200 m	100 m	60 m
Dead end	25 m	18 m	12 m

The following points add clarity and detail to the table above and show factors to be considered when planning and selecting escape routes.

- Dead end travel distances are measured as a person walks.
- Alternative escape routes should be in the opposite direction.
- Semi-open structure: structures where there are substantial openings that would allow smoke and heat from a fire to disperse.
- Low hazard areas: very few flammable materials and the risk of fire occurring is low.
- Medium (normal) hazard areas: include materials that may be flammable. Any fires in this type of area will initially be localised.
- High hazard areas: locations where flammable materials are present in significant quantities. If a fire occurs in this area it will spread rapidly.

During the construction phase escape routes are likely to change and therefore they must be reviewed and alternative routes identified and communicated to everyone on site. Escape routes should be identified at design stage and permanent routes, including fire escape stairs, should be installed as soon as possible. Escape routes must be:

- adequate for the number of workers on site at any one point of the project
- always available in the event of a fire
- kept free from obstructions and checked that they are unobstructed, at least weekly

- altered as necessary as the project advances
- equipped with adequate signage and lighting (old signage must be removed as the escape routes change)
- provided with a second protected escape stairwell for buildings over four storeys high (if not reasonably practicable, due to cost or disruption, an external staircase can be provided instead).

Where an external escape route is constructed against an existing wall, the wall should not have holes or gaps and should provide half an hour of fire resistance for 9 m below the stairway and 1.8 m either side and above (these distances should be measured from the stair treads).

If any opening (such as a window) that is not fire resisting is present, this can be protected with plasterboard, steel sheets or reinforced cement panels.

Doors onto an escape route or the final exit should open outwards (revolving and sliding doors may jam). Doors leading to the escape routes must be self-closing and have half an hour of fire resistance. If doors are required to be locked when the site is unoccupied, due to security reasons, then they must be unlocked as soon as the site is opened.

Preferably all locks should be escape locks that comply with relevant British safety standards, therefore removing the need for a key lock.



Fire doors must comply with BS 476 and their installation must comply with BS 8214.

2.12.6 Emergency lighting and signs

Emergency lighting is required to ensure that escape routes can be seen. In the event of failure of the primary lighting, emergency lighting needs to come on immediately. Batteries or an emergency generator can be used to power the emergency lighting.

Emergency signs should comply with the Health and Safety (Safety Signs and Signals) Regulations 1996. In low light areas or where lighting cannot be guaranteed, photo luminescent escape signs should be provided.

2.13 Firefighting equipment

Portable fire extinguishers, hose reels, hydrants and fire blankets are all types of firefighting equipment.

2.13.1 Portable fire extinguishers

Fire extinguishers do not prevent fires. However, they can be used by trained employees in an attempt to aid escape and minimise loss and damage after a fire has started. It goes without saying that preventing the fire in the first place is a far better option.

Where there is a realistic possibility that staff will have to use fire extinguishers (for example, if they are involved in hot works), they should be trained in their use. CDM (Regulation 32) require that each person at work on a construction site, so far as is reasonably practicable, must be instructed in the correct use of firefighting equipment, which may be necessary for the person to use.

In line with the risks identified in a fire risk assessment, adequate numbers of suitable types of portable fire extinguisher must be provided and kept available throughout the premises.

Extinguishers must be located in clearly visible locations near exits on each floor. They should be fixed to the wall with their carrying handles approximately 1 m above the floor level. Where this is not possible, they should be fixed in position (for example, using base plates or stands) at floor level.

In the open, extinguishers should be situated in red painted boxes, which are either sitting on the floor or raised 500 mm above ground level, with a 'Fire point' sign at a height readily seen above any obstructions. Care must be taken during winter months to ensure that extinguisher contents do not freeze.

To protect electrical distribution panels and items of electrical equipment, appropriate extinguishers (usually carbon dioxide, or CO₂) must be provided near but not dangerously close to the equipment concerned.

For large or costly items of equipment (such as computer suites) consider installing automatic fire detection and extinguishing systems.



All mechanically propelled site plant should carry an appropriate fire extinguisher.

2.13.1.1 Colour coding of fire extinguishers

It is essential that staff who may be called upon to use a fire extinguisher have been trained in their use and have a clear understanding of the colour coding system.

All fire extinguishers are painted red with a contrasting colour somewhere on the casing (such as a contrasting panel or handle), which indicates what extinguishing agent the extinguisher contains.

Colour coding enables a trained person to rapidly identify the type of extinguisher needed in an emergency.

The coloured panels must cover a minimum 5% of the surface area.

FIRE PREVENTION AND CONTROL

Extinguishing medium	Water: for wood, paper, textile and solid material fires	Foam: for flammable liquid fires, such as petrol, diesel, and paint	Powder: for liquid and electrical fires Specialist dry powders: for metal fires	Carbon dioxide (CO₂): for flammable liquid and electrical fires	Wet chemical: for wood, paper, textile, cooking oil and solid material fires
Colour of panel	Red 	Cream 	Blue 	Black 	Yellow 
Where not to use	Do not use on flammable gas and liquids, fires involving electrical equipment or metal fires	Do not use on fires involving electrical equipment or metal fires	Do not use on metal fires unless M28 or L2 text is printed on the extinguisher, which means it is suitable for metal fires	Do not use on metal, wood, paper or textile fires	Do not use on flammable liquid, gas or electrical fires



Dry powder extinguishers may be provided in addition to or substituted for water, foam or CO₂ extinguishers. Extinguishers used to control Class B fires will not work on Class F fires because of the high temperature generated.

A few extinguishers may still be in use that complied with a much earlier standard of colour coding. This standard required that the whole body of each extinguisher was painted in the same contrasting colours that are now used in the later standard. In addition, a small number of extinguishers are bright silver or self-coloured metal with a designated panel stating the medium they contain. The colour coding of the panel should be the same as that listed above.



For further information on the selection and use of fire extinguishers refer to *Fire safety in construction* (HSG168).

2.13.1.2 Inspection and maintenance

Fire extinguishers must be tested by a competent person annually, and there should be an appropriate management system to ensure that they are regularly checked and properly maintained along with other firefighting equipment. These examinations and tests should be in accordance with a recognised procedure, such as that in BS 5306-3:2017 and BS 5306-1:2006. The work should be carried out by a competent person who has received appropriate training. Record the date and results of the examinations, preferably on a service sticker attached to the individual piece of equipment, so that the particular extinguisher or hose reel checked is identifiable.



No extinguisher should be more than 20 years old, have any damage or be unusable in any other way. In that situation, it needs to be replaced or serviced immediately. Clearly, if it has been discharged it will need to be replaced straight away.



Inspection and maintenance

Firefighting equipment must be maintained and regularly inspected. Monthly checks should be carried out to ensure that:

- extinguishers are in place and are suitable for the type of work in the adjacent area
- extinguishers are fully charged and in a good condition, with no visible signs of rust or damage
- all hydrants are clear from obstructions and are clearly marked.

Hose reels and extinguishers must be serviced annually.



Fire extinguishers need to be checked more often if they are in locations where they are likely to be tampered with.

2.13.1.3 Hazardous vapours from fire extinguishers

The discharge of a CO₂ extinguisher in any small, enclosed or confined space will reduce the percentage of oxygen in the air. The dust cloud from a dry powder extinguisher may, in a confined space, produce local and temporary breathing difficulties and poor visibility. Once an extinguisher has been discharged in such circumstances, the user should leave the area immediately. When it is safe to do so, the area should be thoroughly ventilated before allowing anyone to re-enter. If there is a need to re-enter before the air has cleared, suitable breathing apparatus must be worn.

2.13.1.4 Staff training in the selection and use of fire extinguishers

There is a legal requirement for employers to ensure that all employees are familiar with fire procedures, and that an appropriate number of employees are trained in the use of firefighting equipment.

Training in the selection and use of portable extinguishers is essential if they are to be used effectively in fighting small fires. It is also important that the right type of extinguisher is used on certain types of fire.

Attention should be given to the physical strength of persons who may have to use extinguishers. Some extinguishers weigh 20 kg.

During staff training on the use of extinguishers, the following points need to be emphasised.

- Think of evacuation first.
- Only use an extinguisher if it is safe for you to do so.
- Do not let the fire come between you and your escape route. You may become trapped if the fire develops.
- If the extinguisher does not appear to be working or is ineffective on the fire, get out immediately.
- If the fire starts to increase or gets out of control, get out immediately.

During induction training or ongoing staff training, the following points should be made to all staff.

- Do not use a fire extinguisher unless you have been trained to do so.
- Do not misuse fire extinguishers.
- Do not move fire extinguishers from their allocated positions (for example, do not use fire extinguishers as door stops).
- Immediately report any fire extinguishers that appear to have been used, misused or damaged.

2.13.2 Hose reels

Hose reels can be used instead of water-based extinguishers. They are more likely to be found in a live state on projects where premises are being refurbished. One hose reel is recommended for each 800 m² of floor area. In making this assessment, care must be taken to ensure that the hose reaches all of the area to be covered.

On new build projects, or where the water supply to existing hose reels has been drained, it is good practice to make the supply live in progressive stages as the fire loading of the structure increases.

Training employees in the correct use of hose reels is essential if firefighting is to be effective. Staff need to be fully aware of how the hose reels work, and what type of fire they would be effective on, before they use them in an emergency.

2.13.3 Water supplies

Water supplies for firefighting must be available as early as possible. Water can be drawn from the normal mains and hydrants, or from swimming pools and rivers.

As part of the design specification, rising and temporary mains must be provided as the building increases in height, with temporary caps being used, as required, to seal the riser.

Water supplies for firefighting must be tested as the work progresses.

Water supplies for high rise buildings must be agreed with the fire and rescue services before work commences. The on-site water flow should be tested and recorded before work commences and all valves should subsequently be tested every three months.



Fire hydrants must be clearly marked to support identification and be kept clear of obstruction at all times.

2.13.4 Fire blankets

Fire blankets are usually sufficient to deal with small, contained fires involving tea-making facilities, frying pans and so on, and are usually found next to cooking facilities. To fight a small fire using a fire blanket you actually have to approach the fire, so it is important that prior training takes place.

Before attempting to extinguish such fires, the electricity or gas supply must be turned off.

The following points describe how to use a fire blanket.

- The blanket must be pulled from its container with the corners wrapped around the hands of the person using it, making sure that their hands and forearms are completely covered. The blanket must be held at chest level and gently placed over the burning container to exclude the air from the fire.
- The blanket must not be thrown because it may miss the burning container or cause it to spill, spreading the fire.
- The blanket must be left in place until the container has cooled down. It is essential that no-one lifts a corner to check if the fire is out as this may let in enough air to reignite the fire. The container must not be moved before it is cold.

2.14 Temporary accommodation units



In the drive to get the project built, it is easy to overlook the potential fire risks of the site accommodation.

Temporary accommodation units (TAUs) include site offices, canteens, rest rooms, drying rooms and toilets. They are often of timber construction, with fire-resisting surface materials. From site to site they will range from a single site cabin, combining the functions of office and canteen, to a range of single-storey, interlinked units, or they may form a multi-floor administrative complex.

The Confederation of Fire Protection Associations in Europe provides guidance on the fire protection of temporary buildings on construction sites, covering fire risks, fire protection requirements for escape and protection against the spread of fire.



For further information on the recommended frequencies for the inspection and testing of most electrical installations in temporary buildings refer to the Institution of Engineering and Technology.



For further guidance refer to the HSE's *Fire safety in construction* (HSG168).

2.14.1 Location of temporary accommodation units

A fire risk assessment is required for any TAU. The risk assessment will determine the correct location, siting and use of a TAU. They should ideally be located in the open air and separated from the building under construction and storage compounds by a fire break of at least 6 m (20 m on timber-frame and other high-risk sites). If such separation is not possible, the TAU should be designed to meet the following criteria.

- Walls, roof, windows and doors to achieve 30 minutes' fire resistance to BS 476.
- Fire doors must be fitted with self closures.
- Doors and windows should be shut when the area is unoccupied.
- If temporary buildings are to be stacked vertically, the floor or roof assembly and the supporting members must be protected to achieve a minimum of 30 minutes' fire resistance.

For TAUs that are located inside the building under construction, their fire resistance needs more careful consideration and they must include a fitted fire detection system. Under no circumstances should they be located in timber-frame structures. TAUs should be designed and constructed of materials that would not contribute significantly to the growth of a fire or increase levels of smoke or corrosive fumes. Where TAUs do not sit flush on the ground, measures must be taken to prevent the accumulation of rubbish in the space beneath the floor, a gap which will also enhance under floor ventilation.



For an example of a completed fire risk assessment for temporary accommodation units refer to Appendix C.

2.14.2 Sleeping accommodation

Sleeping accommodation could include facilities provided on site or a separate caravan or mobile home site provided solely for the use of the construction project. This accommodation needs a specific fire risk assessment, and sites should aim to provide a fire break at least 10 m wide. If caravans are to be used, they must be spaced in accordance with the Local Authority's requirements, and the necessary fire alarm and firefighting equipment must be provided. There should be no interconnecting route between sleeping accommodation and site.

2.14.3 Fire alarm systems

The fire alarm system for TAUs will inevitably vary between sites. For a single or double site cabin set-up, a verbal shout of 'fire' would be acceptable providing it can be heard throughout the TAU in normal circumstances. Where a fire could remain undetected for a period of time, smoke alarms should be installed.

The more TAUs there are, the more sophisticated the alarm system should be. For small TAU complexes, a hand-operated fire bell might be sufficient, providing it is audible throughout the complex. For larger complexes and areas where cooking is carried out, an electrical break-glass system with multiple call points and sounders may be required. Automatic fire detection should be fitted in canteens and drying rooms.

For TAUs that are within the building under construction, the TAU fire alarm should be integrated with the main building's fire alarm system. Fire alarm systems should be tested weekly to confirm that they are audible throughout the complex. For multiple points, the testing of these points should be rotated. Consideration must be given to fitting automatic fire detection systems and intruder alarms to temporary buildings where flammable substances are stored.



Automatic fire detection must be installed where flammable liquids and gases are stored and in temporary accommodation used for either cooking or for drying clothes.

2.14.4 Means of escape

The site fire safety and evacuation plan should detail the emergency evacuation procedures. In the event of a fire, the first priority is to get everyone out of the TAUs and account for them. How difficult this will be depends upon the size of the TAU complex, how many people are involved and whether they have full mobility. Visitors must also be accounted for.

Toxic smoke is the main cause of fatalities from fires, which makes it a greater hazard in enclosed spaces. Prompt evacuation is paramount. Suitable means of escape from a TAU depends on the travel distance (which determines the position and number of exits and stairs) and the occupancy level (which determines the width of escape routes, including doorways and doors).

For single and double cabin set-up, the means of escape will generally be obvious and special considerations will not be necessary. Ideally, in all TAU complexes there will be escape routes in at least two different directions from all points within the complex.

Where there is only one escape route:

- it must be adequately protected from fire so that it will always be available, if needed
- any high-risk equipment or materials must not be located adjacent to it.

Where the TAU assembly consists of more than three temporary buildings stacked vertically or more than four linked horizontally then specialist advice should be sought from the HSE. Wherever practical, doors should be hung so that they open in the direction of the escape. Employers (or the responsible person) must have regard for the requirements of the Disability Discrimination Act when considering means of escape.



Fire escapes can be blocked unintentionally



For information on fire safety routes and distances refer to the FPA Joint Code of Practice *Fire prevention on construction sites*.

2.14.5 Firefighting equipment

All TAUs should be equipped with firefighting equipment. A hand-held fire extinguisher should be provided as a minimum. The appropriate type of extinguisher will depend upon the class or classes of fire that could occur (refer to 2.13.1 for information on different types of fire extinguisher). A fire blanket will be required in any TAU where cooking oils are heated (such as kitchens and canteens) (refer to 2.13.4).

2.14.6 Other considerations

Emergency lighting should be installed in TAUs if they are to be used outside of normal daylight hours and there is no illumination from other lighting (such as street lighting). Lighting provided should be self-contained or from a separate power source and last for a minimum of one hour.

Heaters in drying rooms and in office spaces must be fixed (preferably at floor level), fitted with security guards and maintained. Heating appliances cover a range of equipment (from wall mounted, electric cabin heaters, to large, forced air LPG heaters). Carelessly drying clothes on heaters can cause fires, so coat stands and drying lockers should be provided at a safe distance from the heaters to prevent the risk of fire. Care must be taken to see that combustible materials are not allowed to build up around such heaters.



For information on carbon monoxide poisoning refer to Chapter B10 Dust and fumes (Respiratory hazards).

Any room where a gas fire is used must be fitted with a permanently open vent or louvre at high and low level to provide sufficient ventilation to enable the fire appliance to operate properly, without producing excess CO₂ (a window that could be closed in cold weather, for example, is not acceptable).

Gas fires, including associated pipework, connections and valves, must be regularly serviced by **Gas Safe registered engineers**. Spare cylinders must not be stored in site cabins or anywhere else where they may be subjected to being heated. All heaters and stoves, including cookers, must be turned off at the end of the working day, with the supply valve on the cylinder closed.

Cooking appliances must be properly installed and have adequate ventilation. To reduce fire risk, provide microwaves instead of electrical or gas cookers. High and low ventilation will also be required, and for heaters and cookers must be permanent and adequate, divided equally between vents at high and low level (for example, a two-burner cooker in a site hut needs approximately 150 mm x 150 mm ventilation. A 3 kW convector heater needs approximately 225 mm x 225 mm ventilation).

TAUs should contain the minimum amount of **furniture and fittings** and, where possible, these should be treated with flame retardants.



Cooking appliances should only be installed and maintained by a competent engineer

Appendix A – Classes of fire

Fires can be placed into one of the following categories.

Class A. Carbonaceous material (such as paper, cloth, wood and rubber) often referred to as solid fuel fires.

Class B. Flammable liquids or liquefiable solids (such as oil, fat, paint and fuel). These can be subdivided into:

- B1.** Fires involving liquids that are soluble in water (such as methanol). They can be extinguished by CO₂, dry powder and water spray
- B2.** Fires involving liquids that are not soluble in water (such as petrol and oil). They can be extinguished using foam, CO₂ and dry powder.

Class C. Flammable gases or liquefied gases (such as propane, butane, hydrogen or acetylene).

Class D. Combustible metals (such as magnesium, sodium and phosphorus).

Class F. High temperature cooking oils or fats (such as those used in deep fat fryers in large catering establishments or restaurants).

Electrical fires (fires involving electrical apparatus or equipment) are not a specific fire class, as electricity is a source of ignition, rather than a fuel. However, they are still important to mention because they have their own special fire safety requirements.

Class A. Carbonaceous material

On the majority of building or construction sites, the following carbonaceous items are freely available sources of fuel.

- Cardboard, paper and cloth.
- Dirty rags, oily rags and clothes.
- Wood.
- Packaging materials.

If a fire occurs involving carbonaceous material, a hose reel or a water extinguisher should be used. The jet of water should be aimed at the base of the fire first, and then moved progressively over the whole of the burning area. Always remove the material from the source of heat if possible, but without endangering the person involved or starting a fire in another location.

Class B. Flammable liquids or liquefiable solids

Includes any fire involving flammable liquids (some examples are shown below).

- Petrol or diesel.
- Paint.
- Oil.
- Resin and adhesive.
- Paraffin.

This type of fire should be dealt with using foam, CO₂ or dry powder extinguishers, depending on whether the fire is contained or flowing.

If the fire is contained, use a foam extinguisher with the jet of foam directed at the back of the container. This allows a blanket of foam to build up and spread across the surface of the burning liquid. If the fire is flowing, a dry powder extinguisher should be directed at the front edge of the fire, in an attempt to separate the flames from the fuel. This will smother the fire, remove oxygen and extinguish it.

The aim of using extinguishers in such a way is for the fire to be covered with a blanket of either foam or dry powder. This will cut off the supply of air, and thus the oxygen, to the fire. Once the blanket has been laid, do not disturb it until the liquid has cooled. Any reintroduction of air may cause the fire to re-ignite.



Never use a water extinguisher or a hose reel on a fire involving any flammable liquid. The water will react violently with the burning liquid and cause an explosion.

Class C. Flammable and liquefied gases

Extreme caution is necessary when dealing with fires involving liquefied gases as there will always be the danger of an explosion.

LPG expands to a ratio of 274:1 so a leak of just one litre of liquid would produce a cloud of gas, if diluted in air to the right concentration, large enough to fill a room 3 m x 2 m x 2 m. This would cause an explosive atmosphere.

If a fire occurs in which a compressed gas cylinder is directly involved the following actions should be taken.

- Call the FRS and tell them the location of the cylinders and type of gas involved.
- Attempt to turn off any gas appliances, if it is safe to do so.
- Attempt to turn the gas off at the cylinder, if it is safe to do so.
- Activate the emergency evacuation procedure and clear the site.

Do not try to fight a fire in which a compressed gas cylinder is directly involved; leave it to the FRS as an overheated cylinder can explode.

If a fire involves other combustible materials (for example, timber that might cause the fire to spread to the location of the cylinders) a decision based upon personal safety will have to be taken on whether to attempt to fight the fire to prevent an escalation of the situation.

Class C fires are best dealt with by the use of dry powder extinguishers.

Class D. Combustible metals

Fires of this type involve magnesium, sodium, phosphorus and similar metals, and should only be dealt with by trained firefighters.



Never apply water to any burning metal fire or a fire involving powdered metal. It would cause an immediate explosive reaction.

Specially-formulated powders are available for use in controlling fire in metals but, as a last resort, if no proprietary powder is available dry sand or earth may be applied to smother the burning area. The proprietary powder should be carefully placed and not thrown onto the burning metal. Throwing the powder will cause the burning material to spread. Be sure either to wear darkened safety glasses or to look away from the extreme brightness whilst attempting to cover the fire. Failure to take these precautions could damage your eyes.

Class F. Cooking oils

Class F fires are fires involving cooking oils or fats. These differ from conventional liquid fires due to the high temperatures involved, with auto ignition temperatures in excess of 340°C. They are difficult to extinguish using conventional extinguishers with a Class B capability, where other flammable liquids (for example, petrol) have a low flash point and so auto ignition temperatures make them easier to extinguish.

Extinguishers designed for cooking oil fires typically include wet chemical, dry chemical, or foam with special additives. These react with the burning oil to create a thick, soapy, heat-resistant crust on top of the cooking oil, preventing flammable vapours from reacting with oxygen. This reaction is called saponification. The alkalinity of the extinguishing material quickly reacts with burning oil to create the soap layer.

Due to the high heat, using an incorrect extinguisher on these fires can be extremely dangerous. Some reasons for this are given below.

- A water jet extinguisher directed at the surface of burning cooking oil will create an explosion. The water is quickly converted into steam, resulting in the possible expulsion of burning oil, which could spread the fire and harm the person using the extinguisher.
- Conventional foam extinguishers can put out flames, but heat quickly destroys the foam blanket, exposing the oil and allowing re-ignition.
- CO₂ and powder extinguishers are effective, but without sealing the oil's surface from oxygen it will rapidly re-ignite.
- Conventional powder, foam or CO₂ extinguishers are normally too powerful and direct, easily splashing the oil and spreading the fire.

Other types of fire

Electrical fires

Electrical equipment fires can be dealt with using CO₂ or dry powder. In staff training, the following simple safety rules should be emphasised.

- Do not use water on any fire involving electrical equipment. You may be electrocuted.
- Switch off the electricity supply, if possible, before fighting the fire. It is then just an ordinary fire.
- Do not approach closer than 1 m to any fire where the electrical supply has not been switched off.
- CO₂ is the best extinguishing medium if the concentration of gas can be confined (for example, within an electrical distribution cupboard).
- Direct the discharge from the extinguisher to one edge of the fire and, with a sweeping movement, pass to the far edge until extinguished.

Electrical equipment used on building and construction sites sometimes incorporates devices to protect against overheating and fire. Most electrical equipment fires are due to misuse or neglect, where appliances have not been properly maintained or are being used for a purpose, or in a manner, for which they were not designed. A fuse larger than the appliance rating will negate the purpose of the fuse and render the appliance potentially unsafe. All staff should be properly trained so that they do not misuse equipment, and to ensure that damaged or defective equipment is reported, taken out of use and professionally repaired.

Flammable adhesives and flammable liquid stores

For such fires, use dry powder or foam. Extreme care must be taken if the adhesive is petroleum or spirit-based, due to explosive vapours.

Cooking ranges

For range fires, use foam, dry powder, CO₂ or a fire blanket. Never move burning pans, as contents may splash and cause serious burns.

Petrol and diesel in stores

For fires involving petrol or diesel in stores use foam or dry powder.

Petrol or diesel-powered plant

For fires involving petrol or diesel-powered plant use foam or dry powder.

Hot work with cutting or welding equipment and bitumen boilers

For fires of this type use foam, dry powder or CO₂. Turn off the heater. The correct types of fire extinguisher must be provided and kept close at hand, with a careful watch being maintained for fire breaking out whilst work is in progress.

Appendix B – Stages of a fire risk assessment

Stage 1. Identify the fire hazards

Anything that burns is fuel for a fire; in general, fires start in one of three ways.

1. Accidentally (for example, failure of a hot work inspection to discover smouldering materials).
2. By act or omission (such as in the example above, if the inspection was simply not carried out).
3. Deliberate arson attack on the structure or, for example, a waste skip.

Therefore, the assessor should consider how a fire could start and what could burn.

Some considerations, relating to the risks, are shown below.

- The use of any equipment that incorporates a naked flame, hot air stream or hot surface.
- Making sure that electrical equipment is not misused (for example, by overloading circuits).
- Grinding and cutting wheel operations (for example, angle grinders or petrol cut-off saws).
- Checking electrical equipment for faults (such as damaged cables or sockets).
- Carrying out periodic testing and inspection of electrical circuits and equipment, including safety devices.
- Discarded smoking materials in areas where smoking is allowed.
- The use of non-fire retardant temporary protection and coverings.
- Buildup of combustible waste, materials and packaging.
- Refuelling in non-designated safe areas.
- Asphalt and bitumen boilers.
- The use of heating and cooking appliances.
- The use of radiant heaters.

For a new-build timber-frame project, it is important to consider these additional areas when compiling the risk assessment.

- The location of the build.
- The proximity to other buildings already in situ.
- Can the materials being used and stored on site be reduced or substituted for less combustible materials?



Potential fire risk due to a build up of materials and packaging near to a radiant heater

Stage 2. Identify the people at risk

Everyone is at risk if there is a fire. Therefore, the assessor should consider the following.

- Decide who are the individuals and/or groups of people likely to be at risk if a fire breaks out and if anyone could be at a greater risk (for example, shift workers, visitors who are unfamiliar with the site, and so on).
- Consider whether anyone is at an enhanced level of risk because of their work activity, or because of a disability (such as a lack of mobility or reduced hearing). The Equality Act requires employers not to discriminate against any person because of their disability and to make reasonable adjustments to match the needs of disabled personnel. A personal emergency evacuation plan (PEEP) may be required in these circumstances.
- Consider the impact that a fire would have on neighbouring properties and all forms of traffic passing the site. For example, the FRS will impose a 200 m cordon and evacuation zone around any fire in which gas cylinders are involved. Such a fire can involve evacuating many people and closing major traffic routes, including railway lines.

Stage 3. Evaluate, remove, reduce and protect from the risk

Having determined the hazards and who is likely to be harmed, the responsible person must take the necessary actions to remove or reduce the risks to an acceptable level.

This is a two-stage process that involves considering the following questions.

1. What are the risks of the fire starting?
2. What are the risks to people on the site and nearby?

There are potentially many practical measures that can be taken to reduce the risk of a fire occurring (listed on the following page).

- Ensuring that electrical office equipment, electrical appliances and portable hand tools are PAT tested at appropriate intervals to identify any defects.
- The effective quarantine of defective electrical equipment awaiting repair (or disposal if not repairable).
- Routinely testing electrical distribution systems.
- Storing and locating flammable and highly flammable liquids in appropriately constructed and secure stores that prevent the escape of spillages or impact by traffic or plant.
- Storing gas cylinders in secure stores that allow any leakage to disperse, with the separation of some gases as is necessary.
- Prohibiting the refuelling of petrol or diesel-powered equipment and plant whilst it is still hot.
- The regular inspection and maintenance as necessary of any system that carries a flammable gas or liquid (such as the supply pipework to oxy-fuel sets, cabin heaters, the canteen cooker or a standby generator).
- Not allowing oxy-fuel sets to be used unless flashback arrestors are fitted to the pipework.
- Preventing arson by putting in place effective security measures, both within and outside of site operating hours.
- The introduction and effective implementation of a hot-works permit system for high-risk activities.
- Ceasing hot work at least one hour (two hours for higher risk sites, such as large, timber-frame projects) before site closure, and regularly monitoring the work area and surrounding structure in the intervening time.
- Maintaining a strict smoking policy; if smoking is permitted in certain areas of the site, provide safe means of disposing of used smoking materials.

Similarly, there are many practical measures that can be taken to reduce the risk to people if a fire should occur.

- Making fire safety and evacuation procedures part of site induction, so site staff know what to do if there is a fire, who will call the emergency services, what escape routes should be used, and so on.
- Understanding how fires spread and what actions to take in the event of a fire.
- Maintaining a strict regime of clearing up waste materials and putting them in skips.
- Ensuring that any flammable liquid waste is stored safely and disposed of at the first opportunity.

Stage 4. Record the findings, prepare an emergency plan and provide training

Record

Where there are five or more employees, the significant findings of the risk assessment, including details of any actions taken to reduce the fire risk, must be recorded in a manner that can be easily retrieved should it be necessary to do so.

Even when there are fewer than five employees, it is good practice to record the **significant** findings of the fire risk assessment. The risk assessment will also assist in developing the emergency plan and will show more clearly where there are weaknesses in fire risk management.

If called upon by the enforcing authority to provide evidence of sound fire risk management practices, a written risk assessment is far preferable to a verbal briefing.

The word **significant** allows the risk assessor to ignore trivial risks of fire. The dividing line between trivial and significant is subjective but it is advisable to err on the side of caution. The risk assessment must be suitable and sufficient for the area it is covering.

Points that require recording include the following.

- Significant fire hazards identified.
- How the hazards will be eliminated.
- Preventative measures that will be taken to reduce the risk of fire from those hazards that cannot be eliminated.
- People who are at risk.
- Actions that have been taken to reduce the spread of fire and smoke if a fire should occur.
- Details of the general fire precautions (such as the maintenance of escape routes, and the location and maintenance of fire-fighting equipment), plus any fire extinguisher training given.
- Details of any instructions given to individuals regarding the actions to take to prevent fires and to escape from the site safely if a fire does occur.
- Details of people with particular fire safety responsibilities (such as the responsible person, fire safety co-ordinator and fire warden).



Record other fire-related issues, including the delivery of appropriate toolbox talks and other fire-related briefings. These records could be useful at a later date.

FIRE PREVENTION AND CONTROL – APPENDIX B

Prepare emergency plan

The findings of the fire risk assessment will help prepare a plan of how the fire can be prevented and how the site staff will be kept safe in the event of a fire. If the site adjoins occupied premises (for example, a hospital or existing business) your plan may need to be co-ordinated with their emergency plan.

Train

All site staff need to know what action to take in the event of a fire and if necessary understand any specific roles they may be required to carry out.

Correct training should be given, as required, and should include the following.

- The significant findings of the fire risk assessment.
- The measures that have been put in place to reduce the risk.
- What people should do in the event of a fire, including temporary staff.
- The identity of people with responsibilities for fire safety (such as fire wardens).
- Any special arrangements that are necessary to protect people on site from serious or imminent danger from fire.

It will also be necessary to provide training on the selection and use of portable fire extinguishers to an appropriate number of staff.

Stage 5. Review and update the fire risk assessment regularly

Due to the ever-changing nature of construction sites, the review of the fire risk assessment is important. A review provides the opportunity to critically examine the effectiveness of the fire risk assessment with regard to site conditions at that time and to update it as necessary. If significant changes in risks are identified or changes are made to the plan, site staff and those in adjoining premises must be informed, as required.

For example, consider the changes that might be necessary as a result of the following.

- Last week's escape route becoming a dead-end today.
- The increasing complexity of escape routes as large structures are built.
- A significant increase in people on site since the last review.
- The start of a hot-work system.
- The start of a new activity that necessitates the storage of flammable substances (such as LPG and other gases) in bulk.
- The need to extend the fire detection system.
- A poorly carried out practice evacuation.
- The occurrence of a fire-related near miss.

Many local FRSs will provide guidance on carrying out fire risk assessments.



Workers should know what to do in the event of a fire

Appendix C – Example of a fire risk assessment for temporary accommodation units

Record of significant findings		
Risk assessment for <i>Single-storey site accommodation comprising five interlinked cabins</i> Site address: <i>Premier Place, Carrow Road, Anytown</i>		Assessment undertaken Date: <i>4th January 2025</i> Completed by: <i>Chris Hughton</i> (Site manager) Signature: <i>C Hughton</i>
Part of temporary accommodation units (TAUs) covered, if not whole area as identified above: <i>N/A</i>		This area is used for: <i>General office work</i> <i>Meeting room and inductions</i> <i>Canteen</i> <i>Toilets, changing and drying room</i>
Step 1. Identify the fire hazards		
Sources of ignition <i>Electrical appliances and office equipment</i> <i>Electrical heater units</i> <i>Cleaning chemicals if mixed</i>	Sources of fuel <i>Rubbish bins</i> <i>Office furniture and equipment</i> <i>Project paperwork and drawings</i>	Sources of oxygen <i>None above ambient level</i>
Step 2. Identify the people at risk		
<i>Office-based and visiting site staff, workers in canteen and toilets and other visitors</i> <i>It is NOT considered likely that the work carried out within the office accommodation could pose any risk of harm to anyone outside the site boundary (12 m away)</i>		
Step 3. Evaluate, remove, reduce and protect from risk		
<i>(3.1) Evaluate the risk of the fire occurring</i> <i>(3.2) Evaluate the risk to people from a fire starting in the premises</i> <i>(3.3) Remove and reduce the hazards that may cause a fire</i> <i>(3.4) Remove and reduce the risks to people from a fire</i>	<i>Meeting room sometimes used for storing used packaging material prior to disposal</i> <i>Desks too close to electric heaters in project office</i> <i>Chairs or clothes too close to electrical heaters in canteen</i> <i>A fire in the meeting room could be undetected for some time</i> <i>A fire in the canteen on ground floor could prevent access to the rear emergency exit</i> <i>Otherwise, emergency evacuation not considered to be problematic</i> <i>Improved housekeeping required to prevent accumulation of flammable waste materials</i> <i>Furniture to be re-arranged – requires relocation of telephone connection point</i> <i>Drying room has enclosed oil-filled covered heaters with hooks for drying clothes – emphasise during induction</i> <i>Fit cages around electrical fires in canteen</i> <i>When the defects noted above have been rectified, the existing fire precautions will be considered to be adequate</i> <i>It is noted that the fire extinguishers are due inspection in early April 2025</i>	
Assessment review		
Review due before <i>2nd August 2025</i>	Completed by	Signature
Review outcome (where substantial changes have occurred a new record sheet should be used)		
Fire action plan		
Will this risk assessment result in a fire action plan being drawn up? YES / NO Delete as applicable		



CONTENTS

Electrical safety



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GT700 Toolbox talks/supporting guidance documents

Toolbox talks on some of these topics are available in the GT700 publication. Supporting guidance documents covering some of these topics are available on our companion website.



Scan the QR code to complete a brief survey, and share your feedback on the course.

Overview

This chapter highlights the dangers associated with working on or near to live electrical circuits, and the good working practices that must be adopted to ensure that such work can be carried out safely.

Each year, accidents involving electricity cause deaths, injuries and burns on construction sites. Furthermore, over the years electricity has been a contributory factor to the starting of several fires on construction sites.

03

3.1 Introduction

Electricity can be regarded as a hidden killer. Unlike many other hazards, it cannot be seen, smelt or heard; it may not become obvious that a cable is live until someone touches it. There may be occasional circumstances where the hum from a transformer or the noise from a motor running indicates that at least some circuits are live, but relying upon this as a risk control measure is certainly not a safe method of working.

A risk of electrocution on site could be created from any of the following.

- Contact or arcing with live parts of the electrical distribution system in any structure being renovated or worked upon, and in associated site accommodation.
- Contact or arcing with overhead power lines.
- Contact or arcing with underground cables.
- The use of defective electrical equipment, including hand tools.

The presence of live electrical cables may not be obvious. Whilst it might be expected that cables run vertically from sockets and switches, this is often not the case and cables may have been run horizontally (and in worst case scenarios, diagonally). These unexpected cable runs, as well as cables that have been run inside partition walls or other hidden voids, need to be given due consideration to avoid accidental damage that can result in serious injury or death.

The temporary electrical distribution systems found on site often operate in a harsh environment and must be treated with care and respect.

The Health and Safety Executive (HSE) regards refurbishment work in buildings as presenting the highest risk. Many of the accidents resulting from such work arise out of the following.

- Confusion and lack of communication as to what circuits have been made dead.
- Hitting cables buried underground, hidden by wall coverings, in voids, and so on.
- Untrained workers knowingly working on live equipment.

Electric shock may not be the only consequence of contact with live electrical circuits; a fault within a circuit or equipment can result in heating or sparking, with the potential to cause a fire or explosion, as well as disrupt services to adjacent properties.



An electrical safety poster should be prominently displayed in several locations, if necessary. It should show basic first-aid procedures for treating someone who has received an electric shock.

There should be somebody on site who is trained in first aid.

3.2 Important points

- No-one other than a qualified electrician should install, maintain or alter the electrical distribution system and the installation must comply with the Electricity at Work Regulations 1989.
- Electrical safety **must** be covered during site induction and refresher training, where it must be emphasised that unauthorised alteration to, or extension of, the site electrical distribution system is strictly prohibited.
- Ideally, apart from mains-powered (230/240 volt) equipment in site offices, all other electrically powered equipment (such as electrical hand tools) should be battery-powered or run off a 110-volt supply.
- The power supply for 230/240 volt equipment must be fed through a safety device in addition to a fuse or circuit breaker (for example, a residual-current device (RCD)). Fuses only protect equipment; they do not 'blow' quickly enough to prevent an electric shock to the user of the equipment.
- The reason behind any safety device tripping, apart from deliberate testing, must be investigated, if necessary by bringing in a qualified electrician; attempts to reset a tripped safety device must not be made until further investigation has been carried out.
- Work on or near to live, exposed cables or equipment poses an obvious danger. Competence is required and such work must only be carried out under a formal **permit to work**.
- Electrical circuits must be tested periodically by a competent person.

- Safety devices (such as RCDs) must be tested frequently by the user in accordance with established guidance.
- Hand-held electrical tools used in construction environments should be subject to a combined inspection and test before first use, and every three months thereafter.
- Electrical equipment must comply with the Electrical Equipment (Safety) Regulations 2016 and have the UKCA marking. If imported from the EU, the product must also show the UKCA mark alongside the CE mark.
- A permit to work system must be introduced and enforced for higher risk work (for example, where a danger from exposed live electrical wires or equipment exists).



For further information on the dangers associated with electricity, refer to Chapter D07 Underground and overhead services.

3.3 Legislative requirements

The regulations are concerned with the safety of electrical installations and equipment and apply to all situations involving electricity, at any place of work. They place legal obligations on duty holders, who may be either an employer or an employee. The details, in the main, are technical and not of general interest to site management staff, providing they have employed competent contractors to carry out electrical work. However, there are some fundamental requirements within the regulations (shown below) which require close observation.

- Working on or near to live electrical parts can only be carried out where:
 - it is unreasonable in all circumstances for the live electrical equipment to be made dead; and
 - it is reasonable in all circumstances for the worker to be at work on or near to the live electrical equipment; and
 - suitable precautions (including, where necessary, the provision of suitable protective equipment) are taken to prevent injury.
- Electrical wiring and equipment, which may be exposed to hostile environments or mechanical damage, should be constructed or protected and/or positioned to avoid danger. Consideration should be given to armoured cabling and conduits.
- All electrical wiring that could result in personal danger must be insulated, otherwise protected or positioned out of harm's way.
- A suitable means must be provided for:
 - cutting off the supply to any item of electrical equipment
 - enabling the supply to be securely isolated
 - identifying all electrical circuits.



Everyone who potentially will come into contact with live electrical wiring or equipment must be fully trained and competent in how to protect themselves and others.

3.3.1 Management of Health and Safety at Work Regulations 1999

These regulations require that employers carry out a suitable and sufficient risk assessment of the work they do and put in place measures to control the risks arising from the work. In the context of this chapter, this will mean that procedures and practices are put in place to eliminate or reduce the risk of anyone being harmed as a result of coming into contact with electricity.



For more information refer to Chapter A05 Risk assessments, method statements and permits to work.

3.3.2 Electricity at Work Regulations 1989

The requirements of these regulations cover fixed and temporary electrical distribution systems rather than electrical hand tools and equipment. The supply system that feeds mains power onto the site and distributes it to site offices, canteens and so on, must be installed, inspected and maintained in accordance with these regulations.

3.3.3 Provision and Use of Work Equipment Regulations 1998 (PUWER)

Electrical equipment and hand tools are classified as *work equipment* as defined under PUWER. As such, employers have a duty to ensure that electrical hand tools and equipment:

- are suitable for their intended purpose
- are well maintained and inspected as necessary
- are fitted with suitable controls
- are fitted with guards as necessary
- can be effectively isolated from the supply
- carry appropriate markings and warning notices.

Furthermore, employers have a duty to ensure that users of work equipment receive adequate training and health and safety information, including written instructions, where appropriate, on the use of the equipment.



For further information on PUWER refer to Chapter C05 Work equipment and hand-held tools.

3.4 Site electrical supply

The supply of electricity on construction sites will ideally be provided from the public supply by the local electricity company. It is generally accepted that temporary supplies installed in line with the *Requirements for electrical installations* (BS 7671) will meet the requirements of the Electricity at Work Regulations. Consideration should also be given to opportunities to use renewable energy sources, where practical.

3.4.1 HSG141 *Electrical safety on construction sites* (second edition)

There has been updated guidance produced by the HSE on the precautions to take to reduce the risk of accidents involving electricity on construction sites. This guidance is aimed at those responsible for planning and subsequent management, and those who control the installation and use of electrical systems and equipment on construction sites. It is divided into three parts, including planning (the pre-construction phase); managing electrical risks (the construction phase); and systems and equipment. The guidance also includes practical information to help understand what the requirements of the relevant legislation may mean in practice, including the Construction (Design and Management) Regulations 2015, the Electricity at Work Regulations and PUWER.

3.4.2 Public supply

A public supply of electricity being provided depends on the following.

- Written application being made to the local electricity company, as soon as possible during the planning stage.
- Name, address and telephone number of the contractor, or principal contractor where appointed, and client, giving the full site address and a location plan.
- Details of the maximum demand load (in kilowatts) that is likely to be required during construction.
- Details of the maximum final demand load (in kilowatts) that will be required when the project is complete and operational.
- Dates when the supply is needed.
- Discussion with electricity company staff to determine the necessary precautions to avoid damage or hazards from any existing overhead or underground cables.
- Establishment of supply points (where incoming cables will terminate), switch gear, metering equipment and requirements for earthing.

3.4.3 Generators

Where a public supply cannot be installed, generators powered by petrol or diesel engines may be used as the main supply. Where a public supply is installed, generators can be used as a back-up supply in emergencies or where the public supply fails. Attention should be given to the siting of generators in order to minimise nuisance caused by noise and fumes. Other environmental issues to mitigate against are fuel spillage and leaks from generators causing water or ground pollution. Consideration should be given to the use of drip trays and bunds.

Site generators range from relatively small portable machines used to power one or two electrical tools to large, permanently sited standby generators capable of powering an extensive distribution system should the public system fail. In the event of unexpected failure, it is normal for standby generating systems to auto-start to avoid any risk associated with a sudden loss of power. It should be noted that some of the smaller machines have dual outputs (230/240 volt and 110 volt). Therefore, separate arrangements for a 1100-volt supply are not necessary. Generally, the output of larger generators is a 400 volt, three-phase supply, aligning with the public supply arrangements.

3.4.5 Calculating power requirements

The calculation for power requirements should be left to an experienced person or to the electricity supply company. To enable the total site requirements to be established, an electrical demand table, which itemises all anticipated electrical loads, must be compiled. It is likely that a member of the site management team will become involved to advise on the electrical loads that will be in use. The person compiling the demand table will apply a diversity factor that acknowledges that not all electrical equipment will be in use at the same time.

3.5 Site distribution systems

On new-build projects, the term *site distribution system* refers to a temporary and flexible electrical system that distributes electrical power around the site for construction purposes. However, with regard to refurbishment work or similar, the distribution may comprise a combination of the fixed electrical system within the building (providing it is safe to leave it live) and elements of a temporary distribution system to enable hand tools, task lighting, and so on to be used.

3.5.1 Safe voltages



Mains voltage (230/240 volts) can be a killer. Battery-powered or 110-volt equipment should always be used, wherever possible.

Generally, the site distribution system comprises of a combination of mains (230/240 volts or 400 volts) and low voltage (110 volts) circuits. To ensure that the appropriate supplies are available where they are required, and that there is no overloading of any part of the system, it must be designed and installed by competent persons. Whilst it is likely that the main incoming supply will be at 230/240 volts or 400 volts, by using a transformer, the voltage will be reduced to a maximum of 110 volts for hand tools, task lighting and so on. In some high-risk situations, transformers are available to provide extra low voltages of 25 or 50 volts.

The way in which 110-volt systems are earthed means that the maximum voltage shock that anyone can receive is around 55 volts, which is regarded as relatively safe for healthy adults. However, this voltage can still induce current flow, which can block the electrical signals between the brain and the muscles. This can stop the heart from beating properly, prevent the person from breathing and cause muscle spasms. Factors that will affect this are the voltage, the parts of the body involved, how damp the person is and the length of time that current is flowing.

3.5.2 Distribution equipment

Electrical distribution equipment must be obtained from a reputable supplier or hire company, to ensure that it has been manufactured to a recognised standard and tested. This is particularly important where electrical distribution equipment is to be sited outdoors, where the ingress of water and dust could be a problem.

There are many types of distribution equipment available, all of which are suited to different situations and environments.

Supply incoming units (SIU) generally have a rating of up to 300 amps per phase. These units include main switch gear and metering equipment.

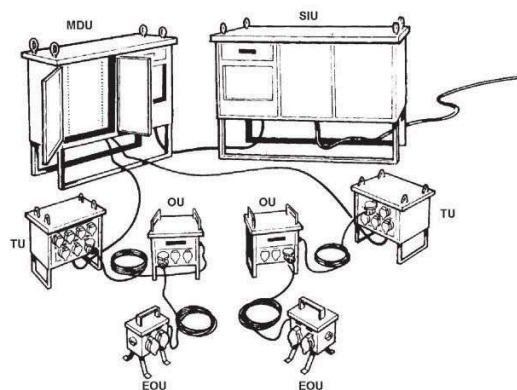
Mains distribution unit (MDU) for the control and distribution of electricity on site, either 400 volts three-phase, or 230/240 volts single-phase AC.

Supply incoming distribution unit (SIDU) may be used in some installations. This is a single piece of equipment that combines the supply and distribution system, which comes as one piece of kit.

Transformer units (TU) are either single-phase (reducing 230/240 volts to 110 volts) or three-phase (reducing 400 volts to 110 volts). Such transformer units can be used for portable tools and plant, and general lighting.

Outlet units (OU) 110-volt socket outlet units rated at 16 or 32 amps. Such outlet units can be used for portable tools, floodlighting and extension outlets. They are not usually protected by circuit-breakers.

Extension outlet units (EOU) equipped with one or more 110-volt socket outlet, rated at 16 amps. Such units can be used for portable tools, local lighting and hand lamps. They are not usually protected by circuit-breakers.



Example of a site electrical distribution system and different types of distribution equipment

In most cases at least part of each piece of distribution equipment will be **colour coded** to indicate its output voltage. Furthermore, the arrangement and number of pins within the electrical connectors are different for each voltage, which prevents, for example, a 110 volt power tool being plugged into a 230/240-volt supply unit.

The colour coding for distribution equipment is shown below.

Violet	25 volts
White	50 volts
Yellow	110–130 volts

Blue	220–240 volts
Red	380–400 volts
Black	500–650 volts



Case study example

On these projects, the site team planned ahead and installed temporary supplies and general lighting to the soffit of main access routes. The lights in the foreground were fixed drops, each with two 110-volt sockets for equipment, which could be moved into rooms as and when required.



3.5.3 Residual-current devices

All attempts should be made to use battery-powered or 110 volt tools where possible. However, it is inevitable that some 230/240-volt equipment will be in use, even if only in the site office. In such circumstances, the 230/240-volt supply must be supplied through an RCD. Under fault conditions, RCDs detect an imbalance in the circuit current and disconnect the supply before the person at risk can receive a potentially fatal electric shock. There are two types of RCD.

1. Those that are fitted as part of an electrical distribution system and are most often found within the supply panel for a site office (this type of RCD can only be installed by a competent electrician). The user should check these daily at the start of every shift, with a more formal inspection every week, and a combined inspection and test before the first use and then every three months.
2. Plug-in RCDs (commonly known as power-breakers) that are plugged into a mains supply socket and have an in-built socket into which an individual mains-powered electrical tool or other appliance is plugged. Plug-in RCDs must undergo a combined inspection and test before the first use and then every month. The user should then check them daily at the start of every shift, with a more formal visual inspection every week.

Both types of RCD have a test button that simulates a fault and operates the device.



The following points should be noted:

- *Maintaining portable electrical equipment* (HSG107) suggests that portable RCDs used on construction sites should have a combined inspection and test before first use.
- RCDs do not reduce current flow or the voltage, only the time that the current flows (about 30 milliseconds), and thereby the severity of the shock.
- RCDs are delicate devices and should be treated with care. Advice on suitable RCDs for construction sites should be sought from manufacturers or suppliers.
- The use of an RCD does not give a 100% guarantee of safety.
- The RCD does not have a fail-safe feature and will not give an indication if it is faulty.

3.5.4 Site accommodation

The incoming electrical supply to site accommodation must be properly designed, installed and commissioned by competent electrical contractors. Site offices and welfare facilities are the only locations where electrical equipment that runs off a 230/240-volt supply should be in use. The electrical supply panel for such facilities must incorporate an RCD in each circuit. The correct operation of each RCD must be confirmed daily by operating the test button, which should trip the device and switch off the circuit running through it.

All portable electrical equipment should be electrical safety tested (portable appliance testing – PAT) at appropriate intervals as decided by a competent person. This includes:

- common types of office equipment (such as scanners and photocopiers)
- kitchen-type equipment (such as kettles, microwave ovens, and so on)
- small items (such as chargers for site radios and battery-powered tools).



For further information refer to the HSE publication *Maintaining portable electrical equipment* (HSG107).

3.6 Safe working practices with electricity

A number of people are killed and injured each year through accidents where electricity was a contributory factor. These accidents occur equally to electricians as to other site operatives. Most fatal accidents involve contact with overhead power lines or unidentified live circuits. Some examples of why these accidents occur are shown below.

- The risks of doing the job had not been adequately assessed.
- Shortcuts in working practices were being taken.
- Pre-work investigation had failed to identify live electrical circuits (particularly in renovation projects).
- Circuits that had been made dead were re-energised by someone who did not realise that the system was being worked on by others.



For further information on working close to overhead cables refer to Chapter D07 Underground and overhead services.



For further details refer to the HSE publications *Electrical safety and you: a brief guide* (INDG231) and *Electricity at work: safe working practices* (HSG85).

3.6.1 Assessing the risks

Working near to live electrical circuits and equipment can only be carried out safely once the significant risks of injury or death have been identified and effective control measures put in place to eliminate the risks or reduce them to an acceptable level. The people doing the work and anyone else who might be adversely affected by it must be informed of the significant risks arising out of the work and the part they must play in ensuring the work can be carried out safely.

Most electrical accidents occur because people are working on or near to equipment that is thought to be dead but which is live, or people are working on equipment known to be live but those carrying out the work do not have adequate training, competency or the correct equipment or adequate resources including, where necessary, accompaniment, to prevent injury.

The person carrying out the risk assessment, and reviewing and updating it at a later stage, must be sufficiently competent to identify the significant risks and to devise effective control measures. The risk assessment must also be reviewed on a regular basis.

3.6.1.1 Accompaniment for authorised live working

It is the senior duty holder's responsibility to determine whether someone carrying out work should be accompanied. This should be based on considerations of how injury is to be prevented. If an accompanying person can substantially contribute towards safe working practice, then they should be present. They should be trained to recognise danger and, if necessary, to render assistance in the event of an emergency.

Below are some examples where a person carrying out electrical work must be accompanied.

- Electrical work involving manipulation of live, uninsulated power conductors at 230 volts or above, using insulated tools.
- Other work on or near to bare live conductors where someone working on their own would not be capable of undertaking the work safely without assistance in, for example, keeping other people away from the work area.

3.6.1.2 Protective equipment

Protective equipment suitable for the work activity should only be used as a last resort (for example, when all other ways to eliminate or reduce risks have been considered). Examples of equipment that can protect someone from the effects of electricity are listed below.

- Suitable clothing, including insulating helmets, goggles and gloves.
- Insulating materials used as fixed or temporary screening to prevent:
 - electric shock
 - short circuit between live conductors and earth.
 - short circuit between live conductors
- Insulating mats and stands to prevent electric shock current via the feet.
- Insulated tools.
- Insulated test probes.

There must be procedures for the periodic inspection, maintenance, examination, testing and/or replacement of this protective equipment.

3.6.2 Managing the risk during refurbishment work

Refurbishment projects pose particular risks if the whole electrical system to the building is not made dead. A number of electrocutions have involved workers who were not electricians but who were carrying out other works (such as plumbers and joiners). The likelihood of these incidents could have been reduced or eliminated by the following.

Understanding the system. Those responsible for planning and managing refurbishment work must understand the electrical system of the building in which the work takes place and liaise with the building occupier. This will enable building work to be planned and managed so that the integrity of the electrical system is not compromised and the workforce remains safe.

System isolation. Relevant parts of the electrical system should be isolated if the refurbishment work (such as joinery or plumbing) is liable to disturb or damage the existing electrical system and expose people to electrical danger.

Portable electrical equipment. Tools, plugs and cables designed for DIY and domestic use are not suitable for site conditions. You should use efficient, robust tools that are battery powered or operate from a 110-volt supply system, earthed so that the maximum voltage does not exceed 55 volts. Regularly inspect power tools and take them out of service if they are damaged. Tools should be serviced by qualified electricians. Do not carry out makeshift repairs.

Protected lighting systems. These protect cabling and bulbs against breakage. If a bulb breaks, the exposed filament may present a hazard. Have a system for checking bulbs regularly to maintain electrical safety and to keep the site well lit.

3.6.3 Effective isolation of circuits

3.6.3.1 Making sure the power is off

If a worker is working next to electrical circuits and is not competent or authorised to check and confirm that the power is off, they must ask a competent person to do it for them, and watch them doing it. If they have any doubts about the method used, they should report the matter to their supervisor immediately.

ELECTRICAL SAFETY

When checking that the power is off the competent person should be **sure** of the following.

- The isolation device being used is suitable for the purpose.
- The isolator being used to turn off the power is working correctly and reliably.
- The switch being isolated is the only way that the circuit can be fed with electrical power.
- The switch being used is **locked** in the off position and cannot be turned on again until the task is complete.
- The equipment and methods being used to check for voltage are in working order and are reliable.
- Isolation has been successful by confirming the circuit is no longer live, using an approved test unit.

Some electrical systems and equipment must be earthed before it is safe to work near them.

Check whether this is necessary, and, if it is, ensure that this is done properly.



Use the lock out, tag out, try out (LOTOTO) procedure.



For further information refer to the Electrical Safety Council guide *Guidance on the management of electrical safety and safe isolation procedures for low voltage installations* (Best practice guide 2).

3.6.3.2 Making sure the power stays off (secure isolation)

If the electrical power has been turned off to allow work to be carried out on an electrical circuit safely, it is essential that the power stays off until the work is finished.

The person doing the work must make sure that **they** are in control and **stay** in control throughout the job. A good way is for that person to have the only key to the switch or to a locked room or cabinet containing the switch.

Removing a fuse may seem to be a safe way of isolating a circuit but someone else could insert another one. Displaying notices is not acceptable, as people often ignore them or do not notice them.

If the person doing the job has any concerns that the electricity may be turned on again without them agreeing, they must be given the authority to stop work without delay and without having to check with a supervisor first.



Electrical systems should be locked out and tagged during any works



Contractor sustains electric shock after failure to isolate power supply

A self-employed contractor sustained 415-volt electric shock injuries from the bare electrical wires supplying an overhead travelling crane while working from a 'cherry picker' installing computer cabling.

The defendant company failed to follow their procedures for safe isolation of the power supply to the crane.

Action

The company was prosecuted under the Health and Safety At Work etc. Act 1974 (Section 3, Sub Section 1) and the Management of Health and Safety at Work Regulations 1992 (No 4, paragraph 1) due to the high risk of serious or fatal injuries from the energised electrical wires. This resulted in a £15,000 fine.

Comment

The company should have undertaken a risk assessment of the cabling activity that should have identified the hazard from the electrical wires. It should have taken action to warn the contractor of the presence of electrical wires near where the work was being carried out.

The company should have taken steps to only allow work when the electrical power had been turned off and to ensure that power remained off for the duration of the work.

(Source: HSE.)

3.6.4 Permit to work

Permit-to-work systems are essential to ensure safe working where live electrical supplies, cables and equipment are present, particularly in installation, maintenance or construction work.

The Electricity at Work Regulations require employers to implement safe systems of work in circumstances where a person could be exposed to live conductors or equipment. A permit to work must be a part of that safe system.

In the case of construction sites, permits to work are more commonly used whilst the site electrical distribution system is itself being worked on, or connections made to it, thereby potentially exposing live parts. Tight control of the permit system is essential if it is to be effective. The person issuing this permit must be competent, and have the appropriate technical knowledge and experience. This is often the electrical contractor's site manager, who must be appointed to raise, issue and co-ordinate permits.

People making the work area and circuit to be worked on safe and the person directly in charge of the work must be involved in completing parts of the permit and signing accordingly. Some companies use permits printed on carbonised paper, which provide identical second or even third copies, where the company's record-keeping system demands them. Most permits contain provisions for the physical locking off circuits, and the retention or display of keys and permits.

Before work starts, the authorised person should ensure that the permit to work will cover the making safe of the part of the system to be worked on, from all possible sources of supply. If the work is handed over from one competent person to another, the permit is to be endorsed by the authorised person and transferred to the second operative. One copy of the permit must be retained for the duration of the work by the person to whom it is issued. Before apparatus is made live again, the permit must be returned for cancellation. Then it must be countersigned by both the holder and the authorised person.

The authorised person should keep a record in the job file of the issued permits and their cancellation.



Under no circumstances should anyone work on, maintain, repair or otherwise adjust high-voltage equipment, except in accordance with the instructions and terms of a formal permit to work.

3.6.5 Locating hidden electrical wiring

The presence of electrical wires may not be obvious at the location where work is to be carried out but, as many people have found out that does not mean they are not there. Even if some wiring is visible, there may be further wiring buried within or hidden by the fabric of the building. Electrical wiring may sometimes look like pipes, and may be a range of colours. The person in charge of the job must complete the following before anyone is allowed to start drilling or cutting into surfaces.

- If feasible, remove the risk by turning off the electricity to the building or area of work.
- Look for the presence of electrical wires and any other hazards (such as asbestos). Remember to look on both sides of walls.
- Inspect plans of the electrical installation, and use these as a guide to finding electrical wiring.
- Use a suitable cable detector if competent, or get a competent person to do it. Remember that some cable detectors will not find a wire carrying a small, or no, current – consult the user guide.
- Identify nearby electrical equipment or installations and try to establish from which direction the wiring runs to them (this will not always be in a horizontal or vertical path).
- Identify and provide equipment that will minimise the risks during the work (for example, cable detectors and insulated tools).
- Ensure that suitable personal protective equipment (PPE) is identified and used.
- Make it clear to the person doing the job that they must **stop work** and check with their supervisor if they have any doubts as to the safety of what they are about to do.



Before a surface is disturbed, a competent person must ensure that a cable and pipeline survey and search are undertaken



An arc flash is an explosive release of energy caused by an electrical short circuit. The temperatures inside an arc flash incident can reach 19,000°C.

3.6.6 Dealing with electric shock

- Could you, or your workers, help someone who has had an electric shock?
- Do you know that what you do could make the difference between life and death?
- Have you and your employees been trained how to apply resuscitation correctly?
- If you were the casualty, is there anybody who could treat you?

ELECTRICAL SAFETY

- Do you have the 'Danger of electric shock' notices available?
- Can these notices or posters be seen by everyone? Are they well positioned?
- Are they read and, more importantly, understood?

3.6.6.1 Electric shock action

If the casualty is in contact with what could be a live electrical supply, switch off the power if possible and shout for help. Never assume the power has been turned off.

If the power cannot be switched off, you must respond in the following way.

- Do not touch the casualty with bare hands.
- If practical and safe to do so, move the casualty away from the electrical source, or the source away from the casualty, with a broom or something else wooden – never use an object which could conduct electricity (such as a metal pole).
- Seek prompt help from a first aider, calling the emergency services immediately to get qualified medical assistance if required.



Electricity can kill. The correct information, instruction, training and supervision help to keep workers and others from coming into contact with unsafe electricity. Never put yourself in danger.

3.6.7 Monitoring contractors

In addition to electrical contractors, many other trades (examples of which are shown below) will have an electrical element to their work.

- Heating and ventilation engineers.
- Lift and escalator engineers.
- Plumbers.
- Fire detection and control engineers.
- Security and closed-circuit television (CCTV) engineers.
- Crane installers.

The work of such contractors and others installing electrical circuits, or simply connecting to electrical equipment installed by others, must be carefully monitored to ensure that they do not jeopardise electrical safety or leave an installation in a dangerous condition.

The **person in charge** of the site, or the appointed responsible person, must complete the following.

- Inspect the risk assessments of these contractors to establish if their work could possibly adversely affect the safety of others working nearby, or those who might otherwise be affected by their work.
- Inspect any written scheme of work (method statement) to establish if any element will conflict with other works.
- Periodically liaise with the contractors to establish if the risk assessment should be reviewed and updated.
- Co-ordinate work, as necessary, to make sure that no-one is put at risk unnecessarily.
- Ensure that higher risk work is carried out in accordance with a permit to work system.
- If there have been any changes on site impacting the original electrical permit-to-work, a new permit must be issued covering the changes. Modifications to the original permit are not permitted.
- Monitor the work of the contractors on a periodic basis to make sure that it is carried out in accordance with the method statement and any permits.
- Make sure that the equipment used by the contractors is suitable for the job in hand, particularly in higher risk situations (such as working in cramped conditions where there is a lot of earthed metalwork or in a damp or wet environment, for example inside a tank).
- Make sure that the contractors implement a strict locking-off procedure where there is potential for them or others to come into contact with live equipment due to circuits being made live again whilst work is still being carried out.
- Where necessary ensure that appropriate signage is displayed to warn others of the dangers, but also appreciate that signage in itself is not an adequate safety measure for isolating a circuit.
- Ensure that contractors do not allow any combustible waste materials to accumulate in areas where it could become the fuel in the event of fire.

Particular care is needed in respect of any alterations and extensions to existing installations (for example, in renovation work), particularly in the identification of circuits where some circuits could remain live for at least part of the work.